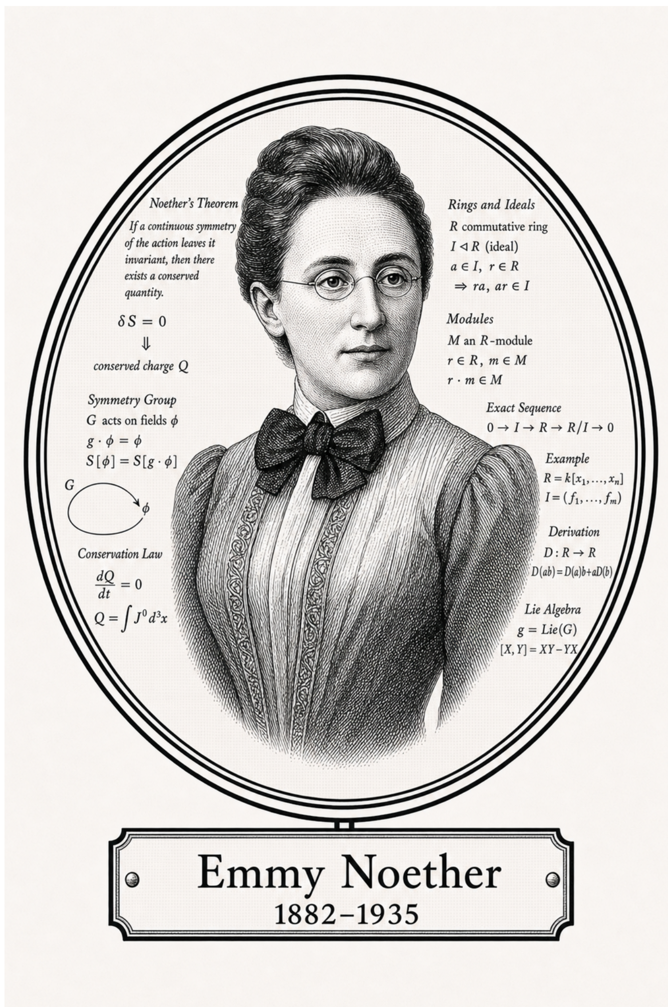
From Cantor to Ito

Algebra



Emmy Noether

1882-1935

Self-Study Notes

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For every reader learning to make mathematics their own.

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Introduction to Algebraic Structures

SERIES
From Cantor to Ito

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CHAPTER 1
Introduction to Algebraic Structures

[

Where You Are in the Journey]

Propositional Logic $\to$ Predicate Calculus $\to$ Sets & Functions $\to$ Proof Techniques $\to$ Real Analysis $\to$ **Intro to Algebraic Structures** $\to$ Linear Algebra $\to$ Topology $\to$ $\cdots$

This chapter fixes the vocabulary of algebraic structure first, then builds the standard one-law and two-law structure tower by adding one axiom at a time.

1.1 What Is a Mathematical Structure?

1.1.1 Algebraic Structures

Remark (Geometric picture). Before any specific algebra, fix what kind of thing we are studying: a set of elements together with stipulated ways to combine them, compare them, and name some of them. The axioms say which stipulations we keep. A group, a ring, and a field are the same general idea with different stipulations.

Definition (Algebraic Structure)

Definition 1.1.1 (Algebraic Structure)**.** An **algebraic structure** of signature $\mathcal{L}$ is a pair

$$\mathcal{A} = (A, I)$$

where A is a nonempty carrier set and I interprets each symbol of $\mathcal{L}$ on A: each n-ary function symbol as an operation $A^n \to A$, each n-ary relation symbol as a subset of A^n, and each constant symbol as an element of A. The structure is algebraic when $\mathcal{L}$ has no relation symbols beyond equality.

Remark (Standard quantified statement).

$$\mathrm{AlgStructure}_{\mathcal{L}}(\mathcal{A}) \iff \mathcal{A} = (A, I),\ A \neq \varnothing,\ \text{and } I \text{ interprets every symbol of } \mathcal{L} \text{ on } A.$$

Remark (Interpretation). The signature is the type; the structure is the instance. "Group" is a signature with a binary operation, an identity constant, and an inverse operation; a particular group interprets these on a carrier.

Remark (Bourbaki anchor). Species of structure, *Theory of Sets*, Chapter IV, Section 1.

Remark (Dependencies).

- Carrier Set
- First-Order Signature

Definition (First-Order Signature)

Definition 1.1.2 (First-Order Signature)**.** A **first-order signature**

$$\mathcal{L} = (\mathrm{Const}_{\mathcal{L}}, \mathrm{Func}_{\mathcal{L}}, \mathrm{Rel}_{\mathcal{L}}, \mathrm{arity})$$

lists constant symbols, function symbols, and relation symbols, with each function or relation symbol carrying a declared arity.

Remark (Dependencies).

- Nullary Operation
- Unary Operation
- Binary Operation

Definition ($\mathcal{L}$-Structure)

Definition 1.1.3 ($\mathcal{L}$-Structure)**.** Given a signature $\mathcal{L}$, an **$\mathcal{L}$-structure** is a nonempty carrier set A together with an interpretation of every symbol of $\mathcal{L}$ on A.

Remark (Dependencies).

- First-Order Signature
- Carrier Set

1.1.2 Laws of Composition

Definition (Binary Operation / Law of Composition)

Definition 1.1.4 (Binary Operation / Law of Composition)**.** Let A be a set. A **binary operation** on A (Bourbaki: an internal law of composition) is a function

$$\star : A \times A \to A.$$

The image of (a, b) is written $a \star b$.

Remark (Standard quantified statement).

$$\mathrm{BinOp}(\star, A) \iff \forall a, b \in A,\ a \star b \in A.$$

Remark (Interpretation). The codomain A is closure: combining two elements never leaves A.

Remark (Bourbaki anchor). "Law of composition," *Algebra I*, Section 1, Definition 1. "Binary operation" is the model-theoretic alias used in this volume.

Remark (Dependencies).

- Function (Volume I recall)
- Carrier Set

Definition (n-ary Operation)

Definition 1.1.5 (n-ary Operation)**.** An **n-ary operation** on A is a function $f : A^n \to A$, with $n \geq 0$. It is unary if $n = 1$, binary if $n = 2$, and nullary if $n = 0$.

Remark (Standard quantified statement).

$$\mathrm{Op}_n(f, A) \iff f : A^n \to A.$$

Definition (Unary Operation)

Definition 1.1.6 (Unary Operation)**.** A **unary operation** on A is a function $f : A \to A$.

Definition (Nullary Operation / Constant)

Definition 1.1.7 (Nullary Operation / Constant)**.** A **nullary operation** on A is the selection of a distinguished element $c \in A$; equivalently, it is an operation $A^0 \to A$.

Remark (Forward reference: external laws). An external law or action $\Omega \times A \to A$, whose elements of Ω are operators, is introduced with actions when modules and vector spaces are reached.

1.1.3 Distinguished Elements

Remark (Geometric picture)*.* Some elements have a privileged role under a law: one that does nothing and one that swallows everything. Defining the one-sided versions first lets the two-sided collapse become a theorem rather than an assumption.

Definition 1.1.8 (Left Identity)**.** An element $e_L \in A$ is a **left identity** for $\star$ if $e_L \star a = a$ for all $a \in A$.

Definition 1.1.9 (Right Identity)**.** An element $e_R \in A$ is a **right identity** for $\star$ if $a \star e_R = a$ for all $a \in A$.

Definition 1.1.10 (Identity / Neutral Element)**.** An element $e \in A$ is a **two-sided identity** (or neutral element) for $\star$ if it is both a left and a right identity:

$$e \star a = a \star e = a \quad \text{for all } a \in A.$$

Remark (Standard quantified statement)*.*

$$\mathrm{Identity}(e, \star, A) \iff e \in A \text{ and } \forall a \in A,\ e \star a = a \star e = a.$$

Proposition 1.1.11 (Identity Collapse and Uniqueness)**.** *If $\star$ has a left identity e_L and a right identity e_R, then $e_L = e_R$. Consequently a two-sided identity, if it exists, is unique.* *Go to proof.*

Remark (Proof structure)*.* Evaluate $e_L \star e_R$ two ways.

Definition 1.1.12 (Left Absorbing Element)**.** An element $z_L \in A$ is **left absorbing** for $\star$ if $z_L \star a = z_L$ for all $a \in A$.

Definition 1.1.13 (Right Absorbing Element)**.** An element $z_R \in A$ is **right absorbing** for $\star$ if $a \star z_R = z_R$ for all $a \in A$.

Definition 1.1.14 (Absorbing Element)**.** An element $z \in A$ is **absorbing** (a zero element) for $\star$ if it is both left absorbing and right absorbing.

Remark (Standard quantified statement)*.*

$$\mathrm{Absorbing}(z, \star, A) \iff z \in A \text{ and } \forall a \in A,\ z \star a = a \star z = z.$$

Proposition 1.1.15 (Uniqueness of the Absorbing Element)**.** *A two-sided absorbing element, if it exists, is unique.* *Go to proof.*

1.1.4 Inverses

Remark (Geometric picture)*.* An inverse undoes an element, returning the identity. Undoing from the left and undoing from the right can a priori differ; associativity forces them to agree.

Definition 1.1.16 (Left Inverse)**.** Let $\star$ have identity e, and let $a \in A$. An element $b \in A$ is a **left inverse** of a if $b \star a = e$.

Definition 1.1.17 (Right Inverse)**.** Let $\star$ have identity e, and let $a \in A$. An element $c \in A$ is a **right inverse** of a if $a \star c = e$.

Definition 1.1.18 (Inverse)**.** Let $\star$ have identity e, and let $a \in A$. An element $a^{-1} \in A$ is an **inverse** of a if

$$a \star a^{-1} = a^{-1} \star a = e.$$

Remark (Standard quantified statement)*.*

$$\text{Inverse}(b, a, \star, e) \iff a \star b = b \star a = e.$$

Definition 1.1.19 (Invertible Element)**.** An element a is **invertible** (a unit) if it has a two-sided inverse.

Remark (Standard quantified statement)*.*

$$\text{Invertible}(a) \iff \exists b,\ a \star b = b \star a = e.$$

Proposition 1.1.20 (Inverse Collapse and Uniqueness in a Monoid)**.** *If $\star$ is associative with identity e, and a has a left inverse b and a right inverse c, then $b = c$. Hence the inverse of an invertible element is unique.* *Go to proof.*

Remark (Proof structure)*.* Compute $b \star a \star c$ two ways; associativity is essential.

Definition 1.1.21 (Additive Inverse)**.** When $\star = +$ with identity 0, the inverse of a is written $-a$ and is called the **additive inverse** or negative.

Definition 1.1.22 (Multiplicative Inverse)**.** When $\star = \cdot$ with identity 1, the inverse of a is written a^{-1} and is called the **multiplicative inverse**.

Proposition 1.1.23 (Units of a Monoid Form a Group)**.** *In a monoid $(A, \star, e)$, the set $A^\times$ of invertible elements is closed under $\star$ and forms a group.* *Go to proof.*

1.1.5 Properties of Laws

Definition 1.1.24 (Associativity)**.** A binary operation $\star$ on A is **associative** if

$$(a \star b) \star c = a \star (b \star c) \quad \text{for all } a, b, c \in A.$$

Remark (Standard quantified statement)*.*

$$\text{Assoc}(\star, A) \iff \forall a, b, c \in A,\ (a \star b) \star c = a \star (b \star c).$$

Theorem 1.1.25 (General Associativity)**.** *If $\star$ is associative, then any finite product $a_1 \star a_2 \star \cdots \star a_n$ has a value independent of parenthesization.* *Go to proof.*

Remark (Proof structure)*.* Induct on n, reducing every bracketing to the right-nested one.

Definition 1.1.26 (Powers / Iterated Operation)**.** For associative $\star$ with identity e, define $a^0 := e$ and $a^{n+1} := a^n \star a$. In additive notation, define $0a := 0$ and $(n+1)a := na + a$.

Definition 1.1.27 (Commutativity)**.** A binary operation $\star$ on A is **commutative** if $a \star b = b \star a$ for all $a, b \in A$.

Definition 1.1.28 (Left Distributivity)**.** For laws $+$ and $\cdot$ on A, multiplication is **left distributive** over addition if

$$a \cdot (b + c) = a \cdot b + a \cdot c$$

for all $a, b, c \in A$.

Definition 1.1.29 (Right Distributivity)**.** For laws $+$ and $\cdot$ on A, multiplication is **right distributive** over addition if

$$(a + b) \cdot c = a \cdot c + b \cdot c$$

for all $a, b, c \in A$.

Definition 1.1.30 (Distributivity)**.** For laws $+$ and $\cdot$ on A, multiplication is **distributive** over addition if it is both left distributive and right distributive.

Remark (Interpretation)*.* Distributivity is the bridge between the two laws of a ring.

Definition 1.1.31 (Idempotent Operation)**.** A binary operation $\star$ on A is **idempotent** if $a \star a = a$ for all $a \in A$.

1.1.6 Regularity Properties

Remark (Geometric picture)*.* A cancellable element is one you can divide out of an equation even when no inverse exists. Where cancellation fails, the witnesses are zero divisors: the obstruction the ring tower is built to remove.

Definition 1.1.32 (Left Cancellable Element)**.** An element $a \in A$ is **left cancellable** if

$$a \star x = a \star y \Longrightarrow x = y$$

for all $x, y \in A$.

Definition 1.1.33 (Right Cancellable Element)**.** An element $a \in A$ is **right cancellable** if

$$x \star a = y \star a \Longrightarrow x = y$$

for all $x, y \in A$.

Definition 1.1.34 (Cancellable Element)**.** An element $a \in A$ is **cancellable** (regular) if it is both left cancellable and right cancellable.

Proposition 1.1.35 (Invertible Implies Cancellable)**.** *In a monoid, every invertible element is cancellable.* *Go to proof.*

Remark (Proof structure)*.* Left-multiply $a \star x = a \star y$ by a^{-1} and use associativity.

Definition 1.1.36 (Zero Divisor)**.** In a ring R, a nonzero element a is a **zero divisor** if there is a nonzero b with $a \cdot b = 0$ or $b \cdot a = 0$.

Remark (Standard quantified statement)*.*

$$\text{ZeroDivisor}(a, R) \Longleftrightarrow a \neq 0 \text{ and } \exists b \neq 0,\ a \cdot b = 0 \text{ or } b \cdot a = 0.$$

Remark (Carried fact)*.* In any ring, 0 is absorbing for multiplication. This is Proposition 1.1.53, derived from distributivity rather than assumed as an axiom.

1.1.7 One Internal Law

Remark (Structure tower rule). Each structure below is the previous structure plus one axiom. Closure is absorbed into the phrase "binary operation."

Definition 1.1.37 (Magma)**.** A **magma** is a pair $(A, \star)$ where $A \neq \varnothing$ and $\star$ is a binary operation on A.

Definition 1.1.38 (Semigroup)**.** A **semigroup** is a magma whose operation is associative.

Definition 1.1.39 (Monoid)**.** A **monoid** is a semigroup with an identity element.

Definition 1.1.40 (Group)**.** A **group** is a monoid in which every element has an inverse.

Definition 1.1.41 (Abelian Group)**.** An **abelian group** is a group whose operation is commutative.

Definition 1.1.42 (Order of a Group)**.** The **order** of a group G, denoted $|G|$, is the cardinality of its carrier. The group is finite if $|G|$ is finite.

Proposition 1.1.43 (Uniqueness of the Identity in a Group)**.** *The identity of a group is unique.* *Go to proof.*

Remark (Dependency). This specializes Proposition 1.1.11.

Proposition 1.1.44 (Uniqueness of Inverses in a Group)**.** *Each element of a group has a unique inverse.* *Go to proof.*

Remark (Dependency). This specializes Proposition 1.1.20.

Proposition 1.1.45 (Cancellation Laws in a Group)**.** *In a group, $ab = ac$ implies $b = c$, and $ba = ca$ implies $b = c$.* *Go to proof.*

Proposition 1.1.46 (Socks–Shoes)**.** *In a group, $(ab)^{-1} = b^{-1}a^{-1}$.* *Go to proof.*

1.1.8 Two Internal Laws

Remark (Convention). Throughout this volume, **ring** means unital ring. A non-unital object with ring-like addition, multiplication, associativity, and distributivity is called a **rng**.

Definition 1.1.47 (Rng)**.** A **rng** is a triple $(R, +, \cdot)$ where $(R, +)$ is an abelian group, $\cdot$ is associative, and $\cdot$ distributes over $+$. No multiplicative identity is required.

Definition 1.1.48 (Ring)**.** A **ring** is a rng with a multiplicative identity 1; equivalently, $(R, \cdot, 1)$ is a monoid. Thus the axioms are: additive abelian group, multiplicative associativity, distributivity, and unity.

Definition 1.1.49 (Commutative Ring)**.** A **commutative ring** is a ring whose multiplication is commutative.

Definition 1.1.50 (Integral Domain)**.** An **integral domain** is a commutative ring with $0 \neq 1$ and no zero divisors.

Definition 1.1.51 (Division Ring / Skew Field)**.** A **division ring** (or skew field) is a ring with $0 \neq 1$ in which every nonzero element is invertible under multiplication. Multiplication is not required to be commutative.

Definition 1.1.52 (Field)**.** A **field** is a commutative division ring; equivalently, it is a commutative ring with $0 \neq 1$ in which $(F \setminus \{0\}, \cdot)$ is a group.

Proposition 1.1.53 (Multiplication by Zero)**.** *In a ring, $a \cdot 0 = 0 \cdot a = 0$ for all a.* *Go to proof.*

Remark (Proof structure)*.* Distribute $a(0+0)$ and cancel additively in $(R, +)$.

Proposition 1.1.54 (Multiplication by Additive Inverse)**.** *In a ring,*

$$a \cdot (-b) = -(a \cdot b) = (-a) \cdot b.$$

In a ring, $(-1) \cdot a = -a$. *Go to proof.*

Proposition 1.1.55 (Cancellation in Integral Domains)**.** *In an integral domain, if $a \neq 0$ and $ab = ac$, then $b = c$.* *Go to proof.*

Proposition 1.1.56 (Every Field is an Integral Domain)**.** *Every field is an integral domain.* *Go to proof.*

Proposition 1.1.57 (Finite Integral Domain is a Field)**.** *Every finite integral domain is a field.* *Go to proof.*

1.1.9 The Structural Lattice

Magma	Rng
$\Downarrow$ add associativity	$\Downarrow$ add unity
Semigroup	Ring
$\Downarrow$ add identity	$\Downarrow$ add multiplicative commutativity
Monoid	Commutative Ring
$\Downarrow$ add inverses	$\Downarrow$ add $0 \neq 1$ and no zero divisors
Group	Integral Domain
$\Downarrow$ add commutativity	$\Downarrow$ finite, or add inverses directly
Abelian Group	Field

Remark (Proof-stub inventory)*.* The one-proof-per-file pipeline for this chapter contains: identity-collapse, absorbing-unique, inverse-collapse, units-form-group, general-associativity, invertible-implies-cancellable, group-identity-unique, group-inverse-unique, group-cancellation, socks-shoes, ring-mult-zero, ring-mult-neg, domain-cancellation, field-is-domain, and finite-domain-is-field.

1.2 Relations

1.3 Operations

1.4 Laws of Operations

1.5 Functions

1.6 Distinguished Elements

1.7 Algebraic Structures

1.8 Groups

1.9 Rings

1.10 Fields

1.11 Number Systems as Structures

1.12 Number Systems as Models

Proofs

Remark (Return). Return to Theorem

Proposition (Identity Collapse and Uniqueness). *If $\star$ has a left identity e_L and a right identity e_R, then $e_L = e_R$. Consequently a two-sided identity, if it exists, is unique.*

Professional Standard Proof.
TODO: professional standard proof for identity-collapse. □

Detailed Learning Proof.
TODO: detailed learning proof for identity-collapse. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Uniqueness of the Absorbing Element). *A two-sided absorbing element, if it exists, is unique.*

Professional Standard Proof.
TODO: professional standard proof for absorbing-unique. □

Detailed Learning Proof.
TODO: detailed learning proof for absorbing-unique. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Inverse Collapse and Uniqueness in a Monoid)**.** *If $\star$ is associative with identity e, and a has a left inverse b and a right inverse c, then $b = c$. Hence the inverse of an invertible element is unique.*

Professional Standard Proof.
TODO: professional standard proof for inverse-collapse. ☐

Detailed Learning Proof.
TODO: detailed learning proof for inverse-collapse. ☐

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Units of a Monoid Form a Group)**.** *In a monoid $(A, \star, e)$, the set $A^\times$ of invertible elements is closed under $\star$ and forms a group.*

Professional Standard Proof.
TODO: professional standard proof for units-form-group. ☐

Detailed Learning Proof.
TODO: detailed learning proof for units-form-group. ☐

Remark (Proof structure)*.*

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (General Associativity). *If $\star$ is associative, then any finite product $a_1 \star a_2 \star \cdots \star a_n$ has a value independent of parenthesization.*

Professional Standard Proof.
TODO: professional standard proof for general-associativity. □

Detailed Learning Proof.
TODO: detailed learning proof for general-associativity. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Invertible Implies Cancellable)**.** *In a monoid, every invertible element is cancellable.*

Professional Standard Proof.
TODO: professional standard proof for invertible-implies-cancellable. □

Detailed Learning Proof.
TODO: detailed learning proof for invertible-implies-cancellable. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Uniqueness of the Identity in a Group)**.** *The identity of a group is unique.*

Professional Standard Proof.
TODO: professional standard proof for group-identity-unique. □

Detailed Learning Proof.
TODO: detailed learning proof for group-identity-unique. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Uniqueness of Inverses in a Group). *Each element of a group has a unique inverse.*

Professional Standard Proof.
TODO: professional standard proof for group-inverse-unique. ☐

Detailed Learning Proof.
TODO: detailed learning proof for group-inverse-unique. ☐

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Cancellation Laws in a Group)**.** *In a group, $ab = ac$ implies $b = c$, and $ba = ca$ implies $b = c$.*

Professional Standard Proof.
TODO: professional standard proof for group-cancellation. □

Detailed Learning Proof.
TODO: detailed learning proof for group-cancellation. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Socks–Shoes). *In a group, $(ab)^{-1} = b^{-1}a^{-1}$.*

Professional Standard Proof.
TODO: professional standard proof for socks-shoes. □

Detailed Learning Proof.
TODO: detailed learning proof for socks-shoes. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Multiplication by Zero)**.** *In a ring, $a \cdot 0 = 0 \cdot a = 0$ for all a.*

Professional Standard Proof.
TODO: professional standard proof for ring-mult-zero. □

Detailed Learning Proof.
TODO: detailed learning proof for ring-mult-zero. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Multiplication by Additive Inverse). *In a ring, $a \cdot (-b) = -(a \cdot b) = (-a) \cdot b$. In a ring, $(-1) \cdot a = -a$.*

Professional Standard Proof.
TODO: professional standard proof for ring-mult-neg. □

Detailed Learning Proof.
TODO: detailed learning proof for ring-mult-neg. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Cancellation in Integral Domains). *In an integral domain, if $a \neq 0$ and $ab = ac$, then $b = c$.*

Professional Standard Proof.
TODO: professional standard proof for domain-cancellation. □

Detailed Learning Proof.
TODO: detailed learning proof for domain-cancellation. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Every Field is an Integral Domain)**.** *Every field is an integral domain.*

Professional Standard Proof.
TODO: professional standard proof for field-is-domain. □

Detailed Learning Proof.
TODO: detailed learning proof for field-is-domain. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Finite Integral Domain is a Field). *Every finite integral domain is a field.*

Professional Standard Proof.
TODO: professional standard proof for finite-domain-is-field. □

Detailed Learning Proof.
TODO: detailed learning proof for finite-domain-is-field. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Cancellation Laws in a Group)**.** *Let G be a group and let $a, b, c \in G$. If $ab = ac$, then $b = c$. If $ba = ca$, then $b = c$.*

Professional Standard Proof.
TODO: professional standard proof for group-cancellation. ☐

Detailed Learning Proof.
TODO: detailed learning proof for group-cancellation. ☐

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Uniqueness of the Identity)**.** *In a group, the identity element is unique.*

Professional Standard Proof.
TODO: professional standard proof for identity-element-unique. □

Detailed Learning Proof.
TODO: detailed learning proof for identity-element-unique. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Uniqueness of Inverses). *In a group, every element has a unique inverse.*

Professional Standard Proof.
TODO: professional standard proof for inverse-element-unique. □

Detailed Learning Proof.
TODO: detailed learning proof for inverse-element-unique. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Cancellation in Integral Domains). *Let R be an integral domain. If $a \neq 0$ and $ab = ac$, then $b = c$.*

Professional Standard Proof.
TODO: professional standard proof for abstract-domain-cancellation. ☐

Detailed Learning Proof.
TODO: detailed learning proof for abstract-domain-cancellation. ☐

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Every Field is an Integral Domain). *Every field is an integral domain.*

Professional Standard Proof.
TODO: professional standard proof for abstract-field-is-domain. □

Detailed Learning Proof.
TODO: detailed learning proof for abstract-field-is-domain. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Cancellation Laws). *Let G be a group and let $a, b, c \in G$. Then:*

(i) ***Left cancellation:*** $ab = ac \implies b = c$.

(ii) ***Right cancellation:*** $ba = ca \implies b = c$.

Professional Standard Proof.
TODO: professional standard proof for group-cancellation-in-groups. □

Detailed Learning Proof.
TODO: detailed learning proof for group-cancellation-in-groups. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Uniqueness of the Identity). *Let G be a group. The identity element of G is unique.*

Professional Standard Proof.
TODO: professional standard proof for group-identity-unique. □

Detailed Learning Proof.
TODO: detailed learning proof for group-identity-unique. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Uniqueness of Inverses)**.** *Let G be a group. For each $a \in G$, the inverse of a is unique.*

Professional Standard Proof.
TODO: professional standard proof for group-inverse-unique. ☐

Detailed Learning Proof.
TODO: detailed learning proof for group-inverse-unique. ☐

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Socks-Shoes Property). *Let G be a group and let $a, b \in G$. Then*

$$(ab)^{-1} = b^{-1}a^{-1}.$$

Professional Standard Proof.
TODO: professional standard proof for group-socks-shoes. □

Detailed Learning Proof.
TODO: detailed learning proof for group-socks-shoes. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Commutativity from Idempotence). *In any Boolean ring R,*

$$pq = qp \quad \textit{for all } p, q \in R.$$

Hence every Boolean ring is a commutative ring.

Professional Standard Proof.
The proof of Proposition **??** established

$$pq + qp = 0 \qquad \text{for all } p, q \in R.$$

Hence $qp = -(pq)$. By Proposition **??**, applied to the element pq, we have $-(pq) = pq$. Therefore $qp = pq$. ☐

Detailed Learning Proof.
The characteristic-2 proof produced a stronger intermediate identity: $pq + qp = 0$. This says that qp is the additive inverse of pq. But in a Boolean ring every element is its own negative, so the additive inverse of pq is just pq itself. Therefore $qp = pq$. ☐

Remark (Proof structure). Reuse the intermediate identity from the characteristic-2 proof, then replace additive inverse by self-negation.

Remark (Dependencies).

- Self Additive Inverse / Characteristic 2.
- Self Negation.
- Commutative Ring.

Remark (Return). Return to Theorem

Corollary (Order is a Power of Two). *A finite Boolean ring has order 2^n for some $n \geq 0$. In particular, there is no Boolean ring of order 3.*

Professional Standard Proof.
By the finite case of Stone representation (Theorem **??**), a finite Boolean ring is isomorphic to $\mathcal{P}(X)$ for some finite set X. If $|X| = n$, then $|\mathcal{P}(X)| = 2^n$. Since 3 is not a power of 2, no Boolean ring has order 3. □

Detailed Learning Proof.
Finite Boolean rings are set rings in disguise. Stone representation says that every finite one looks like the power set of a finite set. A set with n elements has exactly 2^n subsets, so the ring has 2^n elements. The number 3 is not on that list. □

Remark (Proof structure). Invoke the finite Stone representation and count subsets.

Remark (Dependencies).

- Stone Representation for Boolean Rings.
- Cardinality of a finite power set.

Remark (Return). Return to Theorem

Proposition (Self Additive Inverse / Characteristic 2)**.** *In any Boolean ring R,*

$$p + p = 0 \quad \textit{for every } p \in R.$$

Professional Standard Proof.
Let $p, q \in R$. By B1 the element $p + q$ is idempotent, so

$$(p + q)(p + q) = p + q. \tag{$*$}$$

Expand the left side by distributivity (R3), applied on both sides:

$$(p + q)(p + q) = p(p + q) + q(p + q) = pp + pq + qp + qq.$$

Apply B1 to $pp = p$ and $qq = q$, and regroup by additive associativity (R1):

$$p + q = p + pq + qp + q.$$

Cancel p and q using additive cancellation (Proposition 1.1.45):

$$pq + qp = 0 \qquad \text{for all } p, q \in R. \tag{$**$}$$

Specialize $(**)$ to $q = p$ and apply B1 once more:

$$p + p = pp + pp = 0.$$

□

Detailed Learning Proof.
The only new fact over a plain ring is that everything squares to itself. Apply it to a sum, because squaring a sum is where addition and multiplication are forced to interact.

Square $p + q$ two ways: it equals itself by idempotence, and it expands to the four products pp, pq, qp, qq by distributivity. There is no $2pq$ shortcut because multiplication is not assumed commutative.

Idempotence collapses pp to p and qq to q, leaving $p + pq + qp + q$. Match this against $p + q$ and cancel the shared p and q: this leaves $pq + qp = 0$.

Put $q = p$. Idempotence turns $pp + pp$ into $p + p$, so $p + p = 0$. Characteristic 2 was never assumed; it is forced by idempotence. □

Remark (Proof structure). Use B1 on $(p + q)^2$, expand by R3, cancel the outer terms by R1, and specialize the resulting identity to $q = p$.

Remark (Dependencies).

- Boolean Ring (B1).
- Ring axioms R1 and R3.
- Additive cancellation.
- Lean reference: `BooleanAlgebraDefinitions.lean`.

Remark (Return). Return to Theorem

Proposition (Self Negation)**.** *In any Boolean ring R,*

$$-p = p \quad \textit{for every } p \in R.$$

Professional Standard Proof.
By Proposition **??**, $p+p=0$. By the additive-inverse clause of R1, $p+(-p)=0$. Both p and $-p$ are therefore additive inverses of p; by uniqueness of additive inverses (Proposition 1.1.44), $-p=p$. □

Detailed Learning Proof.
An additive inverse of p is anything that can be added to p to reach 0. The ring gives one such element, $-p$. Proposition **??** hands us a second: p itself, since $p+p=0$. Inverses in an abelian group are unique, so the two candidates are the same element: $-p=p$. The minus sign is therefore pure decoration in a Boolean ring. □

Remark (Proof structure)*.* Use characteristic 2 to make p an additive inverse of itself, then use uniqueness of additive inverses.

Remark (Dependencies).

- Self Additive Inverse / Characteristic 2.
- Ring axiom R1.
- Uniqueness of additive inverses.

Remark (Return). Return to Theorem

Proposition (Cancellation in Integral Domains). *Let R be an integral domain and $a, b, c \in R$ with $a \neq 0$. Then*

$$ab = ac \implies b = c.$$

Professional Standard Proof.
TODO: professional standard proof for domain-cancellation. □

Detailed Learning Proof.
TODO: detailed learning proof for domain-cancellation. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Multiplication by Additive Inverse)**.** *Let R be a ring. For all $a, b \in R$,*

$$a \cdot (-b) = -(a \cdot b) \quad \textit{and} \quad (-a) \cdot b = -(a \cdot b).$$

In any ring with unity, $(-1) \cdot a = -a$.

Professional Standard Proof.
TODO: professional standard proof for ring-mult-neg. □

Detailed Learning Proof.
TODO: detailed learning proof for ring-mult-neg. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Multiplication by Zero)**.** *Let R be a ring. For all $a \in R$,*

$$a \cdot 0 = 0 \cdot a = 0.$$

Professional Standard Proof.
TODO: professional standard proof for ring-mult-zero. ☐

Detailed Learning Proof.
TODO: detailed learning proof for ring-mult-zero. ☐

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Stone Representation for Boolean Rings)**.** *Every Boolean ring is isomorphic to a field of sets. Every finite Boolean ring is isomorphic to $\mathcal{P}(X)$ for some finite set X, hence has $2^{|X|}$ elements.*

Professional Standard Proof.
TODO: professional standard proof for stone-representation-boolean-rings. ☐

Detailed Learning Proof.
TODO: detailed learning proof for stone-representation-boolean-rings. ☐

Remark (Proof structure).

Remark (Dependencies).

- Boolean Ring.
- TODO: prime ideals or ultrafilters.

Remark (Return). Return to Theorem

Proposition (Characteristic of a Field). *Let $\mathbb{F}$ be a field. The* characteristic *of $\mathbb{F}$ is the smallest positive integer n such that*

$$\underbrace{1 + 1 + \cdots + 1}_{n} = 0,$$

or 0 if no such n exists. The characteristic of a field is either 0 or a prime p.

Professional Standard Proof.
TODO: professional standard proof for field-characteristic. □

Detailed Learning Proof.
TODO: detailed learning proof for field-characteristic. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Nonzero Scalars Have Inverses)**.** *Let $\mathbb{F}$ be a field and $a \in \mathbb{F}$ with $a \neq 0$. Then there exists a unique $a^{-1} \in \mathbb{F}$ such that $a \cdot a^{-1} = 1$.*

Professional Standard Proof.
TODO: professional standard proof for field-inverse-exists. □

Detailed Learning Proof.
TODO: detailed learning proof for field-inverse-exists. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Every Field is an Integral Domain)**.** *Every field is an integral domain.*

Professional Standard Proof.
TODO: professional standard proof for field-is-domain. □

Detailed Learning Proof.
TODO: detailed learning proof for field-is-domain. □

Remark (Proof structure)*.*

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Zero Product Property in a Field). *Let $\mathbb{F}$ be a field and $a, b \in \mathbb{F}$. Then*

$$ab = 0 \implies a = 0 \ \textit{ or } \ b = 0.$$

Professional Standard Proof.
TODO: professional standard proof for field-zero-product. ☐

Detailed Learning Proof.
TODO: detailed learning proof for field-zero-product. ☐

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Capstone

Chapter 2

Abstract Algebra

SERIES
From Cantor to Ito

VOLUME VI
Algebra

BOOK I
Algebra

CHAPTER 2
Abstract Algebra

2.1 Induction

2.1.1 Induction

Definitions and Theorems

2.2 Integers

2.2.1 Integers

Basic Definitions and Theorems

2.3 Modular Arithmetic

2.3.1 Modular Arithmetic

Basic Definitions and Theorems

Proofs

Capstone

Chapter 3

Algebraic Geometry

3.1 Algebraic Geometry

3.1.1 Preliminary Definitions

Definition 3.1.1 (Ring)**.** A set R with two binary operations $+ : R \times R \to R$ and $\cdot : R \times R \to R$ is a *ring* if:

1. **Additive structure:** $(R, +)$ is an abelian group.
2. **Associativity of multiplication:**
$$(a \cdot b) \cdot c = a \cdot (b \cdot c) \quad \text{for all } a, b, c \in R.$$
3. **Multiplicative identity:** There exists $1 \in R$ such that
$$1 \cdot a = a \cdot 1 = a \quad \text{for all } a \in R.$$
4. **Distributivity:**
$$a \cdot (b + c) = a \cdot b + a \cdot c \quad \text{and} \quad (a + b) \cdot c = a \cdot c + b \cdot c \quad \text{for all } a, b, c \in R.$$

Remark (Standard quantified statement)*.* A ring is a set equipped with addition and multiplication satisfying the additive-group, multiplicative-associativity, multiplicative-identity, and distributive laws.

Remark (Interpretation)*.* A ring generalizes a field by dropping the requirement that nonzero elements have multiplicative inverses, and by not requiring multiplication to be commutative. When multiplication is commutative, we call R a *commutative ring*.

Definition 3.1.2 (Commutative Ring)**.** A ring R is *commutative* if

$$a \cdot b = b \cdot a \quad \text{for all } a, b \in R.$$

Remark (Standard quantified statement)*.* A ring R is commutative when multiplication in R is commutative.

Remark (Structural Summary)*.* The hierarchy of algebraic structures encountered so far is:

Structure	Additive Group	Multiplicative Structure
Ring	Abelian	Associative, with identity
Commutative Ring	Abelian	Associative, commutative, with identity
Field	Abelian	Abelian on nonzero elements

Every field is a commutative ring, but not every commutative ring is a field.

Polynomial Rings

Definition 3.1.3 (Polynomial)**.** Let R be a commutative ring. A *polynomial* in one variable over R is a formal expression

$$f = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0,$$

where $n \in \mathbb{N}$, the *coefficients* $a_0, a_1, \ldots, a_n \in R$, and x is a formal symbol called an *indeterminate*.

Remark (Standard quantified statement)*.* A polynomial over R is a finite formal sum $\sum_i a_i x^i$ with coefficients $a_i \in R$.

Remark (Interpretation)*.* The indeterminate x is not a variable ranging over values in R. It is a formal placeholder that encodes the coefficient sequence. Two polynomials are equal if and only if all their coefficients are equal.

Definition 3.1.4 (Degree)**.** Let $f = a_n x^n + \cdots + a_0$ be a polynomial over R. If $a_n \neq 0$, then the *degree* of f is

$$\deg(f) := n.$$

The coefficient a_n is called the *leading coefficient* of f. A polynomial with leading coefficient 1 is called *monic*.

Remark (Standard quantified statement)*.* For a nonzero polynomial $f = a_n x^n + \cdots + a_0$ with leading coefficient $a_n \neq 0$, the degree of f is n.

Remark (Interpretation)*.* The zero polynomial $f = 0$ has no leading coefficient. Its degree is left undefined, or assigned $-\infty$ by convention to preserve the identity $\deg(fg) = \deg(f) + \deg(g)$.

Definition 3.1.5 (Polynomial Ring)**.** Let R be a commutative ring. The *polynomial ring* over R in one indeterminate x, denoted $R[x]$, is the set of all polynomials in x with coefficients in R, equipped with addition and multiplication defined by:

$$\left(\sum_i a_i x^i\right) + \left(\sum_i b_i x^i\right) := \sum_i (a_i + b_i) x^i,$$

$$\left(\sum_i a_i x^i\right) \cdot \left(\sum_j b_j x^j\right) := \sum_k \left(\sum_{i+j=k} a_i b_j\right) x^k.$$

Remark (Standard quantified statement)*.* The set $R[x]$ is the collection of finite formal sums in x with coefficients in R, equipped with coefficientwise addition and Cauchy-product multiplication.

Example (Polynomial arithmetic over the integers)*.* Let $R = \mathbb{Z}$ and consider

$$f = 2x^2 + 3x + 1, \qquad g = x + 4 \quad \in \mathbb{Z}[x].$$

Then:

$$f + g = 2x^2 + 4x + 5,$$
$$f \cdot g = 2x^3 + 11x^2 + 13x + 4.$$

Remark (Structural Summary)*.* With these operations, $R[x]$ is itself a commutative ring. The original ring R embeds into $R[x]$ as the constant polynomials.

Property	Holds in $R[x]$?
Commutative ring	Always
Field	Only in degenerate cases
R embeds in $R[x]$	Always

Polynomial Rings in Several Variables

Definition 3.1.6 (Polynomial Ring in n Variables)**.** Let R be a commutative ring and let $n \in \mathbb{N}$. The *polynomial ring in n variables* over R, denoted

$$R[x_1, x_2, \ldots, x_n],$$

is defined inductively by

$$R[x_1, \ldots, x_n] := R[x_1, \ldots, x_{n-1}][x_n].$$

Its elements are finite sums of the form

$$f = \sum_\alpha a_\alpha \, x^\alpha,$$

where the sum ranges over multi-indices $\alpha = (\alpha_1, \ldots, \alpha_n) \in \mathbb{N}^n$, the coefficients $a_\alpha \in R$, and $x^\alpha := x_1^{\alpha_1} \cdots x_n^{\alpha_n}$.

Remark (Standard quantified statement). The ring $R[x_1, \ldots, x_n]$ consists of finite formal sums $\sum_\alpha a_\alpha x^\alpha$ indexed by multi-indices.

Definition 3.1.7 (Monomial). A *monomial* in $R[x_1, \ldots, x_n]$ is a polynomial of the form

$$x^\alpha = x_1^{\alpha_1} \cdots x_n^{\alpha_n}$$

for some multi-index $\alpha \in \mathbb{N}^n$.

Remark (Standard quantified statement). A monomial in $R[x_1, \ldots, x_n]$ is a single multi-index power x^α.

Definition 3.1.8 (Total Degree). The *total degree* of the monomial x^α is

$$|\alpha| := \alpha_1 + \alpha_2 + \cdots + \alpha_n.$$

The degree of a polynomial $f \in R[x_1, \ldots, x_n]$ is the maximum total degree among all monomials with nonzero coefficient.

Remark (Standard quantified statement). The total degree of x^α is $|\alpha| = \alpha_1 + \cdots + \alpha_n$.

Example (Example). In $\mathbb{R}[x, y, z]$, the polynomial

$$f = 3x^2y + xy^2z - 5z^3$$

has three terms with total degrees 3, 4, and 3 respectively. Hence $\deg(f) = 4$.

Remark (Relevance to Algebraic Geometry). Polynomial rings in several variables are the foundational algebraic object of algebraic geometry. The geometric objects studied in subsequent sections — affine varieties, ideals, coordinate rings — are all defined in terms of $k[x_1, \ldots, x_n]$ for a field k. The interplay between the algebra of $k[x_1, \ldots, x_n]$ and the geometry of its zero sets is the central theme of Clader–Ross.

Ideals

Definition 3.1.9 (Ideal). Let R be a commutative ring. A subset $I \subseteq R$ is an *ideal* of R if:

1. $0 \in I$,
2. $a + b \in I$ for all $a, b \in I$,
3. $r \cdot a \in I$ for all $r \in R$ and $a \in I$.

Remark (Standard quantified statement). An ideal of R is an additive subgroup of R closed under multiplication by arbitrary elements of R.

Definition 3.1.10 (Generated Ideal). Let R be a commutative ring and let $f_1, \ldots, f_m \in R$. The *ideal generated by* $f_1, \ldots, f_m$ is

$$\langle f_1, \ldots, f_m \rangle := \left\{ \sum_{i=1}^{m} r_i f_i : r_i \in R \right\}.$$

An ideal of this form is called *finitely generated*.

Remark (Standard quantified statement). The ideal generated by $f_1, \dots, f_m$ is the set of all finite R-linear combinations of those generators.

Definition 3.1.11 (Finitely Generated Ideal)**.** An ideal $I \subseteq R$ is *finitely generated* if there exist $f_1, \dots, f_m \in R$ such that $I = \langle f_1, \dots, f_m \rangle$.

Remark (Standard quantified statement). An ideal I is finitely generated when $I = \langle f_1, \dots, f_m \rangle$ for some finite list $f_1, \dots, f_m$ of elements of R.

Definition 3.1.12 (Noetherian Ring)**.** A commutative ring R is *Noetherian* if every ideal of R is finitely generated.

Remark (Standard quantified statement). A commutative ring R is Noetherian when every ideal $I \subseteq R$ is finitely generated.

Theorem 3.1.13 (Hilbert Basis Theorem)**.** *If R is a Noetherian ring, then the polynomial ring $R[x]$ is also Noetherian.*

In particular, $k[x_1, \dots, x_n]$ is Noetherian for any field k. *Go to proof.*

Remark (Standard quantified statement). If R is Noetherian, then $R[x]$ is Noetherian.

Remark (Interpretation). The Hilbert Basis Theorem guarantees that every ideal in $k[x_1, \dots, x_n]$ is finitely generated. This is a foundational finiteness result: it means every algebraic variety can be cut out by finitely many polynomial equations.

Quotient Rings

Definition 3.1.14 (Quotient Ring)**.** Let R be a commutative ring and let $I \subseteq R$ be an ideal. The *quotient ring* R/I is the set of cosets

$$R/I := \{a + I : a \in R\},$$

equipped with addition and multiplication defined by

$$(a + I) + (b + I) := (a + b) + I,$$
$$(a + I) \cdot (b + I) := (a \cdot b) + I.$$

Remark (Standard quantified statement). For an ideal I of R, the quotient R/I is the set of additive cosets of I with operations induced from R.

Remark (Interpretation). The quotient ring R/I is a commutative ring. Elements of R/I are equivalence classes under the relation $a \sim b \iff a - b \in I$.

Example (A complex quotient). In $\mathbb{R}[x]$, consider the ideal $I = \langle x^2 + 1 \rangle$. The quotient ring

$$\mathbb{R}[x]/\langle x^2 + 1 \rangle \;\cong\; \mathbb{C},$$

since in the quotient, x satisfies $x^2 = -1$, which is precisely the defining relation of $i \in \mathbb{C}$.

Remark (Structural Position). Quotient rings of polynomial rings are the coordinate rings of affine varieties, which are studied in depth in the next chapter. The passage

$$k[x_1, \dots, x_n] \;\longrightarrow\; k[x_1, \dots, x_n]/I$$

is the algebraic encoding of restricting from all of $\mathbb{A}^n$ to the variety $V(I)$.

Proofs

Remark (Return). Return to Theorem

Theorem (Hilbert Basis Theorem). *If R is a Noetherian ring, then the polynomial ring $R[x]$ is also Noetherian.*

In particular, $k[x_1, \ldots, x_n]$ is Noetherian for any field k.

Professional Standard Proof.
TODO: professional standard proof for hilbert-basis-theorem. □

Detailed Learning Proof.
TODO: detailed learning proof for hilbert-basis-theorem. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Capstone

Chapter 4

Linear Algebra

4.1 Vector Space Preliminaries

Remark (Toolkit — Vector Space Preliminaries). **Concept family:** The concrete foundations for $\mathbf{F}^n$ before the abstract vector space axioms are introduced.

Atomic items in this section:

- Definition: List
- Definition: $\mathbf{F}^n$
- Definition: Addition in $\mathbf{F}^n$
- Theorem: Addition Is Commutative in $\mathbf{F}^n$
- Definition: Zero Vector in $\mathbf{F}^n$
- Definition: Additive Inverse in $\mathbf{F}^n$
- Definition: Scalar Multiplication in $\mathbf{F}^n$

Key predicate:

$$\text{ListEquality}((x_1, \dots, x_n), (y_1, \dots, y_m)) \;:\equiv\; n = m \;\wedge\; \forall k \leq n \; (x_k = y_k).$$

Definition (List)

Definition 4.1.1 (List)**.** Suppose n is a non-negative integer. A **list** of length n is an ordered collection of n elements.

Remark (Standard quantified statement).

$$\text{List}(L) \iff \exists n \in \mathbb{N} \cup \{0\} : \big(\text{Length}(L) = n \wedge L = (x_1, x_2, \ldots, x_n)\big)$$

Remark (Definition predicate reading).

$$\text{List}(L) \iff \exists n \in \mathbb{N}_0, \exists (x_1, \ldots, x_n) : L = (x_1, \ldots, x_n)$$

Remark (Interpretation). A list is the primary mechanism for discussing ordered finite sets of vectors or scalars. Unlike a set, the structural integrity of a list depends on both the identity and the position of its members.

- **Order Sensitivity**: The lists (a, b) and (b, a) are distinct unless $a = b$.
- **Finiteness**: By Axler's convention, a list must have a finite number of elements. Infinite ordered collections are categorized as sequences.
- **Empty List**: A list of length $n = 0$ is called the empty list, denoted by ().

Remark (Dependencies).

- Function
- Whole Numbers

Definition ($\mathbf{F}^n$)

Definition 4.1.2 ($\mathbf{F}^n$). Let $\mathbf{F}$ denote $\mathbb{R}$ or $\mathbb{C}$, and let n be a positive integer. Then $\mathbf{F}^n$ is defined to be the set of all lists of length n of elements of $\mathbf{F}$:

$$\mathbf{F}^n = \{(x_1, \ldots, x_n) : x_1, \ldots, x_n \in \mathbf{F}\}.$$

The entries in a list in $\mathbf{F}^n$ are called the coordinates of the list.

Remark (Standard quantified statement).

$$\begin{aligned} &x \in \mathbf{F}^n \\ &\quad \iff \quad \exists x_1, \ldots, x_n \in \mathbf{F} \text{ such that } x = (x_1, \ldots, x_n). \end{aligned}$$

Remark (Definition predicate reading).

$$\text{Fn}(\mathbf{F}, n) \;:\equiv\; \mathbf{F}^n = \{(x_1, \ldots, x_n) : x_k \in \mathbf{F}\}.$$

Remark (Interpretation). An element of $\mathbf{F}^n$ is an ordered list of n scalars from $\mathbf{F}$. The order matters, and the list is completely specified by its coordinates.

Remark (Dependencies).

- Definition: List

Definition (Addition in $\mathbf{F}^n$)

Definition 4.1.3 (Addition in $\mathbf{F}^n$)**.** Let

$$x = (x_1, \ldots, x_n), \qquad y = (y_1, \ldots, y_n)$$

be vectors in $\mathbf{F}^n$. The sum of x and y is defined by

$$x + y = (x_1 + y_1, \ldots, x_n + y_n).$$

Remark (Standard quantified statement)*.*

$$\begin{aligned} &\forall x, y \in \mathbf{F}^n, \\ &\quad x = (x_1, \ldots, x_n) \wedge y = (y_1, \ldots, y_n) \Rightarrow x + y = (x_1 + y_1, \ldots, x_n + y_n). \end{aligned}$$

Remark (Interpretation)*.* Addition in $\mathbf{F}^n$ is defined coordinatewise: add the first coordinates, then the second coordinates, and so on through the nth coordinates.

Remark (Dependencies)*.*

- Definition: $\mathbf{F}^n$

Theorem (Addition Is Commutative in $\mathbf{F}^n$)

Theorem 4.1.4 (Addition Is Commutative in $\mathbf{F}^n$)**.** *For all $x, y \in \mathbf{F}^n$,*

$$x + y = y + x.$$

Go to proof.

Remark (Standard quantified statement)*.*

$$\forall x, y \in \mathbf{F}^n,\ x + y = y + x.$$

Remark (Interpretation)*.* Because addition in $\mathbf{F}^n$ is coordinatewise and scalar addition in $\mathbf{F}$ is commutative, vector addition in $\mathbf{F}^n$ is commutative.

Remark (Dependencies)*.*

- Definition: Addition in $\mathbf{F}^n$

Definition (Zero Vector in $\mathbf{F}^n$)

Definition 4.1.5 (Zero Vector in $\mathbf{F}^n$)**.** The symbol 0 denotes the list of length n all of whose coordinates are 0:

$$0 = (0, \ldots, 0) \in \mathbf{F}^n.$$

Remark (Standard quantified statement)*.*

$$0 = (0, \ldots, 0) \in \mathbf{F}^n.$$

Remark (Interpretation). The zero vector is the vector whose every coordinate is the scalar 0 in $\mathbf{F}$. It is the additive identity for $\mathbf{F}^n$.

Remark (Dependencies).

- Definition: $\mathbf{F}^n$

Definition (Additive Inverse in $\mathbf{F}^n$)

Definition 4.1.6 (Additive Inverse in $\mathbf{F}^n$)**.** Let

$$x = (x_1, \dots, x_n) \in \mathbf{F}^n.$$

The additive inverse of x is defined by

$$-x = (-x_1, \dots, -x_n).$$

Remark (Standard quantified statement).

$$\begin{aligned} &\forall x \in \mathbf{F}^n, \\ &\quad x = (x_1, \dots, x_n) \;\Rightarrow\; -x = (-x_1, \dots, -x_n). \end{aligned}$$

Remark (Interpretation). The additive inverse in $\mathbf{F}^n$ is formed coordinatewise by taking the additive inverse of each coordinate in $\mathbf{F}$.

Remark (Dependencies).

- Definition: $\mathbf{F}^n$
- Definition: Zero Vector in $\mathbf{F}^n$

Definition (Scalar Multiplication in $\mathbf{F}^n$)

Definition 4.1.7 (Scalar Multiplication in $\mathbf{F}^n$)**.** Let

$$\lambda \in \mathbf{F} \quad \text{and} \quad x = (x_1, \dots, x_n) \in \mathbf{F}^n.$$

The product of λ and x is defined by

$$\lambda x = (\lambda x_1, \dots, \lambda x_n).$$

Remark (Standard quantified statement).

$$\begin{aligned} &\forall \lambda \in \mathbf{F}\ \forall x \in \mathbf{F}^n, \\ &\quad x = (x_1, \dots, x_n) \;\Rightarrow\; \lambda x = (\lambda x_1, \dots, \lambda x_n). \end{aligned}$$

Remark (Interpretation). Scalar multiplication in $\mathbf{F}^n$ is defined coordinatewise: multiply each coordinate of the vector by the scalar λ.

Remark (Dependencies).

- Definition: $\mathbf{F}^n$

Remark (Toolkit — Vector Spaces). **Concept family:** The abstract vector space axioms and their immediate consequences.

Atomic items in this section:

- Definition: Vector Space (10 axioms)
- Definition: Vector and Point
- Definition: Real Vector Space
- Definition: Complex Vector Space
- Theorem: Unique Additive Identity
- Theorem: Unique Additive Inverse
- Theorem: $0v = 0$
- Theorem: $a0 = 0$
- Theorem: $(-1)v = -v$

Key predicate:

$$\mathrm{VectorSpace}(V, \mathbf{F}, +, \cdot).$$

The 10 axioms are closure, commutativity, associativity, additive identity, additive inverse, scalar closure, both distributive laws, scalar associativity, and the scalar identity.

Definition (Vector Space)

Definition 4.1.8 (Vector Space)**.** A **vector space** over $\mathbf{F}$ is a set V together with an addition on V and a scalar multiplication

$$\mathbf{F} \times V \to V$$

such that the following properties hold for all $u, v, w \in V$ and all $a, b \in \mathbf{F}$:

(i) $u + v \in V$;

(ii) $u + v = v + u$;

(iii) $(u + v) + w = u + (v + w)$;

(iv) there exists $0 \in V$ such that $v + 0 = v$;

(v) for every $v \in V$ there exists $-v \in V$ such that $v + (-v) = 0$;

(vi) $av \in V$;

(vii) $a(u + v) = au + av$;

(viii) $(a + b)v = av + bv$;

(ix) $a(bv) = (ab)v$;

(x) $1v = v$.

Remark (Standard quantified statement).

$$\begin{aligned} &V \text{ is a vector space over } \mathbf{F} \\ &\iff \quad V \text{ is closed under addition and scalar multiplication,} \\ &\quad \text{addition on } V \text{ is commutative and associative,} \\ &\quad V \text{ has an additive identity and additive inverses,} \\ &\quad \text{and scalar multiplication satisfies the two distributive laws,} \\ &\quad \text{the associative law for scalar multiplication, and the identity law.} \end{aligned}$$

Remark (Definition predicate reading).

$$\text{VectorSpace}(V, \mathbf{F}, +, \cdot)$$

Remark (Negated quantified statement).

$$\neg \text{ VectorSpace}(V, \mathbf{F}, +, \cdot) \iff \text{at least one of the ten axioms fails for some } u, v, w \in V,\ a, b \in \mathbf{F}. \blacksquare$$

Remark (Interpretation). A vector space is a set whose elements can be added together and multiplied by scalars, with these operations satisfying the natural algebraic rules familiar from $\mathbf{F}^n$.

Remark (Dependencies).

- Definition: $\mathbf{F}^n$ (motivating example)
- Definition: Addition in $\mathbf{F}^n$
- Definition: Scalar Multiplication in $\mathbf{F}^n$

Definition (Vector and Point)

Definition 4.1.9 (Vector and Point)**.** An element of a vector space V is called a **vector**. When $V = \mathbf{F}^n$, an element of $\mathbf{F}^n$ may also be called a **point**.

Remark (Standard quantified statement).

$$v \in V \implies v \text{ is a vector},$$
$$x \in \mathbf{F}^n \implies x \text{ is a point of } \mathbf{F}^n.$$

Remark (Interpretation). The word *vector* emphasizes the algebraic role of an element of V. In $\mathbf{F}^n$, the same object can also be viewed geometrically as a point with coordinates in $\mathbf{F}$.

Remark (Dependencies).

- Definition: Vector Space
- The Real Numbers

Definition (Real Vector Space)

Definition 4.1.10 (Real Vector Space)**.** A **real vector space** is a vector space over $\mathbb{R}$.

Remark (Standard quantified statement).

$$V \text{ is a real vector space} \iff V \text{ is a vector space over } \mathbb{R}.$$

Remark (Definition predicate reading).

$$\mathrm{RealVectorSpace}(V) \;:\equiv\; \mathrm{VectorSpace}(V, \mathbb{R}, +, \cdot).$$

Remark (Interpretation). In a real vector space, the scalars used in scalar multiplication are real numbers.

Remark (Dependencies).

- Definition: Vector Space
- The Real Numbers

Definition (Complex Vector Space)

Definition 4.1.11 (Complex Vector Space)**.** A **complex vector space** is a vector space over $\mathbb{C}$.

Remark (Standard quantified statement).

$$V \text{ is a complex vector space} \iff V \text{ is a vector space over } \mathbb{C}.$$

Remark (Definition predicate reading).

$$\text{ComplexVectorSpace}(V) \;:\equiv\; \text{VectorSpace}(V, \mathbb{C}, +, \cdot).$$

Remark (Interpretation). In a complex vector space, the scalars used in scalar multiplication are complex numbers.

Remark (Notation). For each positive integer m, the notation $\mathbf{F}^m$ denotes the vector space of all lists of length m of elements of $\mathbf{F}$.

The notation $\mathbf{F}^\infty$ denotes the set of all infinite lists

$$(x_1, x_2, x_3, \dots)$$

with each $x_j \in \mathbf{F}$, equipped with coordinatewise addition and coordinatewise scalar multiplication.

Remark (Dependencies).

- Definition: Vector Space

Remark (Coordinates in Vector Spaces). A pointwise proof is available when vectors are given explicitly by coordinates, as in $\mathbf{F}^n$, because the operations are defined coordinatewise. In an arbitrary vector space, vectors need not come with coordinates, so proofs must use the vector space axioms rather than coordinate calculations.

Theorem (Unique Additive Identity)

Theorem 4.1.12 (Unique Additive Identity)**.** *A vector space has a unique additive identity.* *Go to proof.*

Remark (Standard quantified statement).

$$\begin{aligned} &\text{If } 0, 0' \in V \text{ each satisfy} \\ &\qquad \forall v \in V,\ v + 0 = v \quad \text{and} \quad \forall v \in V,\ v + 0' = v, \\ &\text{then } 0 = 0'. \end{aligned}$$

Remark (Interpretation). Although the vector space axioms assert the existence of an additive identity, they force that identity to be unique.

Remark (Dependencies).

- Definition: Vector Space

Theorem (Unique Additive Inverse)

Theorem 4.1.13 (Unique Additive Inverse)**.** *Every vector in a vector space has a unique additive inverse.* *Go to proof.*

Remark (Standard quantified statement).

$$\forall v \in V,\ \text{if } w, u \in V \text{ satisfy } v + w = 0 \text{ and } v + u = 0,$$
$$\text{then } w = u.$$

Remark (Interpretation). The vector space axioms guarantee that each vector has an additive inverse, and that inverse is determined uniquely by the requirement that the sum be 0.

Remark (Notation). If $v \in V$, then the unique additive inverse of v is denoted by $-v$.

If $w, v \in V$, then

$$w - v \ := \ w + (-v).$$

Remark (Dependencies).

- Definition: Vector Space
- Theorem: Unique Additive Identity

Theorem ($0v = 0$)

Theorem 4.1.14 ($0v = 0$)**.** *For every $v \in V$,*

$$0v = 0.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall v \in V,\ 0v = 0.$$

Remark (Interpretation). Multiplying any vector by the scalar 0 gives the zero vector. The 0 on the left is the scalar zero in $\mathbf{F}$; the 0 on the right is the zero vector in V.

Remark (Dependencies).

- Definition: Vector Space
- Theorem: Unique Additive Inverse

Theorem ($a0 = 0$)

Theorem 4.1.15 ($a0 = 0$)**.** *For every $a \in \mathbf{F}$,*

$$a0 = 0.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall a \in \mathbf{F},\ a0 = 0.$$

Remark (Interpretation). Every scalar sends the zero vector to the zero vector. The 0 on the right of a is the zero vector in V.

Remark (Dependencies).

- Definition: Vector Space
- Theorem: Unique Additive Inverse

Theorem $((-1)v = -v)$

Theorem 4.1.16 $((-1)v = -v)$**.** *For every* $v \in V$*,*

$$(-1)v = -v.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall v \in V,\ (-1)v = -v.$$

Remark (Interpretation). Scalar multiplication by -1 produces the additive inverse of a vector. This justifies the notation $-v$ for the additive inverse: it is literally (-1) times v.

Remark (Dependencies).

- Definition: Vector Space
- Theorem: Unique Additive Inverse
- Theorem: $0v = 0$

Remark (Toolkit — Vector Subspaces). **Concept family:** Subspaces, their sums, and direct sums.

Atomic items in this section:

- Definition: Subspace
- Theorem: Conditions for a Subspace
- Definition: Sum of Subspaces
- Theorem: Sum of Subspaces Is the Smallest Containing Subspace
- Definition: Direct Sum
- Theorem: Condition for a Direct Sum
- Theorem: Direct Sum of Two Subspaces

Key predicates:

Subspace(U, V), ClosedUnderAddition(U), ClosedUnderScalarMultiplication$(U, \mathbf{F})$, DirectSu

Key test: U is a subspace $\iff$ $0 \in U$, U closed under $+$, U closed under scalar multiplication.

Definition (Subspace)

Definition 4.1.17 (Subspace)**.** Let V be a vector space over $\mathbf{F}$. A subset $U \subseteq V$ is called a **subspace** of V if U is itself a vector space over $\mathbf{F}$ with the same addition and scalar multiplication as on V.

Remark (Standard quantified statement).

$$\begin{aligned} &U \text{ is a subspace of } V \\ &\iff \quad U \subseteq V \text{ and } U \text{ is a vector space over } \mathbf{F} \\ &\quad \text{under the operations inherited from } V. \end{aligned}$$

Remark (Definition predicate reading).

$$\text{Subspace}(U, V) \;:\equiv\; U \subseteq V \;\wedge\; \text{VectorSpace}(U, \mathbf{F}, +, \cdot).$$

Remark (Negated quantified statement).

$$\neg\, \text{Subspace}(U, V) \iff U \not\subseteq V \;\vee\; \neg\, \text{VectorSpace}(U, \mathbf{F}, +, \cdot).$$

Remark (Interpretation). A subspace is a subset of a vector space that is closed under the vector-space operations and therefore forms a vector space in its own right.

Remark (Dependencies).

- Definition: Vector Space

Theorem (Conditions for a Subspace)

Theorem 4.1.18 (Conditions for a Subspace)**.** *Let V be a vector space over $\mathbf{F}$ and let $U \subseteq V$. Then U is a subspace of V if and only if the following three conditions hold:*

(i) $0 \in U$*;*

(ii) for all $u, v \in U$*, one has* $u + v \in U$*;*

(iii) for all $a \in \mathbf{F}$ *and all* $u \in U$*, one has* $au \in U$*.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned} &U \text{ is a subspace of } V \\ &\iff \quad 0 \in U \\ &\quad \wedge\ \text{ClosedUnderAddition}(U) \\ &\quad \wedge\ \text{ClosedUnderScalarMultiplication}(U, \mathbf{F}). \end{aligned}$$

Remark (Failure modes). U fails to be a subspace via exactly one of three routes: it omits the zero vector, it contains two vectors whose sum falls outside U, or it contains a vector and a scalar whose product falls outside U.

Remark (Contrapositive quantified statement).

$$\neg\ \mathrm{Subspace}(U, V) \iff 0 \notin U \lor \neg\ \mathrm{ClosedUnderAddition}(U) \lor \neg\ \mathrm{ClosedUnderScalarMultiplication}(U, \mathbf{F}).$$

Remark (Interpretation). To check that a subset is a subspace, it suffices to verify the three conditions. The remaining seven vector space axioms are inherited automatically from V since U uses the same operations. The three conditions are minimal: condition (i) ensures U is non-empty; (ii) and (iii) ensure the inherited operations stay within U.

Remark (Dependencies).

- Definition: Subspace
- Definition: Vector Space

Definition (Sum of Subspaces)

Definition 4.1.19 (Sum of Subspaces)**.** Let V be a vector space over $\mathbf{F}$, and let $U_1, \ldots, U_m$ be subspaces of V. The **sum** of $U_1, \ldots, U_m$ is the set

$$U_1 + \cdots + U_m = \{u_1 + \cdots + u_m : u_1 \in U_1, \ldots, u_m \in U_m\}.$$

Remark (Standard quantified statement).

$$\begin{aligned} &v \in U_1 + \cdots + U_m \\ &\quad \iff \quad \exists u_1 \in U_1\ \cdots\ \exists u_m \in U_m \text{ such that } v = u_1 + \cdots + u_m. \end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{SumOfSubspaces}(W, \{U_1, \ldots, U_m\}, V) \ :\equiv\ W = U_1 + \cdots + U_m.$$

Remark (Interpretation). The sum of subspaces consists of all vectors that can be built by adding together one vector from each subspace. Vectors from different subspaces interact; the representation $v = u_1 + \cdots + u_m$ need not be unique.

Remark (Dependencies).

- Definition: Subspace

Theorem (Sum of Subspaces Is the Smallest Containing Subspace)

Theorem 4.1.20 (Sum of Subspaces Is the Smallest Containing Subspace)**.** *Let V be a vector space over $\mathbf{F}$, and let $U_1, \ldots, U_m$ be subspaces of V. Then $U_1 + \cdots + U_m$ is a subspace of V containing each of $U_1, \ldots, U_m$. Moreover, if W is a subspace of V containing each of $U_1, \ldots, U_m$, then*

$$U_1 + \cdots + U_m \subseteq W.$$

Go to proof.

Remark (Standard quantified statement).

$$
\begin{aligned}
&\mathrm{Subspace}(U_1 + \cdots + U_m,\, V),\\
&U_j \subseteq U_1 + \cdots + U_m \quad \text{for } j = 1, \dots, m,\\
&\forall W \subseteq V \ \Big[\big(W \text{ is a subspace of } V \wedge U_1 \subseteq W \wedge \cdots \wedge U_m \subseteq W\big)\\
&\qquad\qquad \Rightarrow\ U_1 + \cdots + U_m \subseteq W\Big].
\end{aligned}
$$

Remark (Interpretation). The sum is the smallest subspace containing all the given subspaces: it is their join in the lattice of subspaces of V. Any subspace containing all of $U_1, \dots, U_m$ must contain every sum $u_1 + \cdots + u_m$, so it contains the sum subspace.

Remark (Dependencies).

- Definition: Sum of Subspaces
- Theorem: Conditions for a Subspace

Definition (Direct Sum)

Definition 4.1.21 (Direct Sum)**.** Let V be a vector space over $\mathbf{F}$, and let $U_1, \dots, U_m$ be subspaces of V such that

$$V = U_1 + \cdots + U_m.$$

Then V is called the **direct sum** of $U_1, \dots, U_m$ if each vector $v \in V$ can be written in the form

$$v = u_1 + \cdots + u_m, \qquad u_1 \in U_1, \dots, u_m \in U_m,$$

in exactly one way. In this case, we write

$$V = U_1 \oplus \cdots \oplus U_m.$$

Remark (Standard quantified statement).

$$
\begin{aligned}
&V = U_1 \oplus \cdots \oplus U_m\\
&\quad \iff \quad V = U_1 + \cdots + U_m\\
&\qquad \wedge\ \forall v \in V\ \exists!\, u_1 \in U_1\ \cdots\ \exists!\, u_m \in U_m \text{ such that } v = u_1 + \cdots + u_m.
\end{aligned}
$$

Remark (Definition predicate reading).

$$\mathrm{DirectSum}(V, \{U_1, \dots, U_m\}, \mathbf{F}) \ :\equiv\ V = U_1 \oplus \cdots \oplus U_m.$$

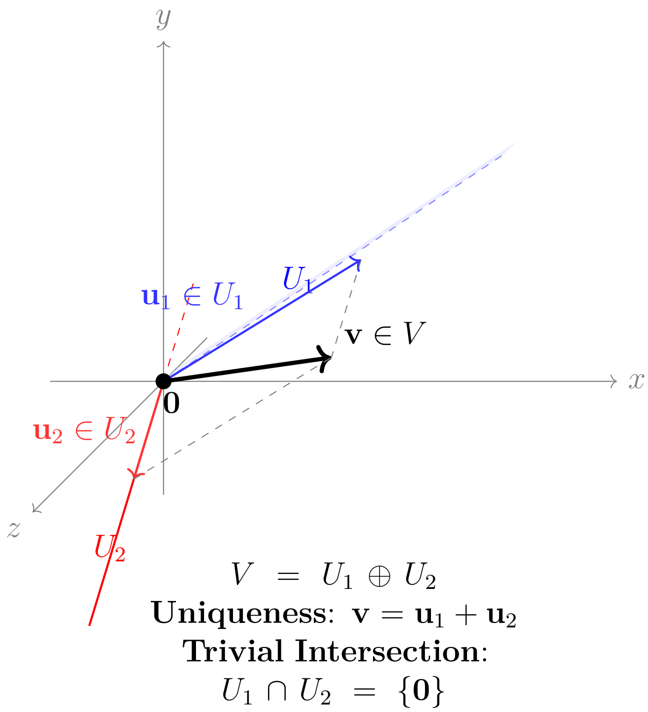
Remark (Negated quantified statement).

$$\neg\,\mathrm{DirectSum}(V, \{U_1, \dots, U_m\}, \mathbf{F}) \iff \exists v \in V\ \exists \text{ two distinct decompositions of } v.$$

Remark (Interpretation). A direct sum is a sum in which every vector has a unique decomposition into pieces coming from the given subspaces. The uniqueness condition is the structural rigidity that makes direct sums so useful — it means the subspaces contribute independently, with no redundancy.

Remark (Dependencies).

- Definition: Sum of Subspaces



Theorem (Condition for a Direct Sum)

Theorem 4.1.22 (Condition for a Direct Sum)**.** *Let V be a vector space over* $\mathbf{F}$*, and let $U_1, \ldots, U_m$ be subspaces of V such that*

$$V = U_1 + \cdots + U_m.$$

Then the following are equivalent:

(i) $V = U_1 \oplus \cdots \oplus U_m$*;*

(ii) if $u_1 \in U_1, \ldots, u_m \in U_m$ and

$$u_1 + \cdots + u_m = 0,$$

then

$$u_1 = \cdots = u_m = 0.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned} &V = U_1 \oplus \cdots \oplus U_m \\ &\quad \Longleftrightarrow \quad \forall u_1 \in U_1 \ \cdots \ \forall u_m \in U_m \\ &\qquad \big(u_1 + \cdots + u_m = 0 \Rightarrow u_1 = \cdots = u_m = 0\big). \end{aligned}$$

Remark (Interpretation). A sum is direct exactly when the zero vector has only the trivial decomposition as a sum of vectors from the given subspaces. This is the practical test: rather than verifying uniqueness for every $v \in V$, it suffices to check uniqueness at $v = 0$.

Remark (Dependencies).

- Definition: Direct Sum
- Theorem: Unique Additive Identity

Theorem (Direct Sum of Two Subspaces)

Theorem 4.1.23 (Direct Sum of Two Subspaces)**.** *Let U and W be subspaces of a vector space V. Then*

$$U + W = U \oplus W$$

if and only if

$$U \cap W = \{0\}.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned} &U + W = U \oplus W \\ &\iff \quad U \cap W = \{0\}. \end{aligned}$$

Remark (Contrapositive quantified statement).

$$U \cap W \neq \{0\} \iff U + W \neq U \oplus W.$$

If the two subspaces share a nonzero vector, the sum is not direct.

Remark (Interpretation). For two subspaces, directness is equivalent to saying they overlap only in the zero vector. This gives a simple geometric test: draw both subspaces and check whether their intersection is just the origin. The result does not generalise to three or more subspaces — it is possible for $U_1 \cap U_2$, $U_1 \cap U_3$, and $U_2 \cap U_3$ to all equal $\{0\}$ yet $U_1 + U_2 + U_3$ still fail to be a direct sum.

Remark (Dependencies).

- Theorem: Condition for a Direct Sum
- Definition: Direct Sum

Proofs

Remark (Return). Return to Theorem

Theorem (Addition Is Commutative in F^n)**.** *For all* $x, y \in \mathbf{F}^n$,

$$x + y = y + x.$$

Professional Standard Proof.
TODO: professional standard proof for addition-commutative-in-fn. ☐

Detailed Learning Proof.
TODO: detailed learning proof for addition-commutative-in-fn. ☐

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Unique Additive Identity). *A vector space has a unique additive identity.*

Professional Standard Proof.
TODO: professional standard proof for unique-additive-identity. □

Detailed Learning Proof.
TODO: detailed learning proof for unique-additive-identity. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Unique Additive Inverse)**.** *Every vector in a vector space has a unique additive inverse.*

Professional Standard Proof.
TODO: professional standard proof for unique-additive-inverse. □

Detailed Learning Proof.
TODO: detailed learning proof for unique-additive-inverse. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (0v = 0). *For every $v \in V$,*

$$0v = 0.$$

Professional Standard Proof.
TODO: professional standard proof for zero-scalar-times-vector. □

Detailed Learning Proof.
TODO: detailed learning proof for zero-scalar-times-vector. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem ($a0 = 0$)**.** *For every $a \in \mathbf{F}$,*

$$a0 = 0.$$

Professional Standard Proof.
TODO: professional standard proof for scalar-times-zero-vector. □

Detailed Learning Proof.
TODO: detailed learning proof for scalar-times-zero-vector. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem ($(-1)v = -v$)**.** *For every $v \in V$,*

$$(-1)v = -v.$$

Professional Standard Proof.
TODO: professional standard proof for minus-one-times-vector. ☐

Detailed Learning Proof.
TODO: detailed learning proof for minus-one-times-vector. ☐

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Conditions for a Subspace)**.** *Let V be a vector space over $\mathbf{F}$ and let $U \subseteq V$. Then U is a subspace of V if and only if the following three conditions hold:*

(i) $0 \in U$;

(ii) for all $u, v \in U$, *one has* $u + v \in U$;

(iii) for all $a \in \mathbf{F}$ *and all* $u \in U$, *one has* $au \in U$.

Professional Standard Proof.
TODO: professional standard proof for conditions-for-a-subspace. □

Detailed Learning Proof.
TODO: detailed learning proof for conditions-for-a-subspace. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Sum of Subspaces Is the Smallest Containing Subspace)**.** *Let V be a vector space over $\mathbf{F}$, and let $U_1, \ldots, U_m$ be subspaces of V. Then $U_1 + \cdots + U_m$ is a subspace of V containing each of $U_1, \ldots, U_m$. Moreover, if W is a subspace of V containing each of $U_1, \ldots, U_m$, then*

$$U_1 + \cdots + U_m \subseteq W.$$

Professional Standard Proof.
TODO: professional standard proof for sum-of-subspaces-smallest-containing. □

Detailed Learning Proof.
TODO: detailed learning proof for sum-of-subspaces-smallest-containing. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Condition for a Direct Sum)**.** *Let V be a vector space over $\mathbf{F}$, and let $U_1, \dots, U_m$ be subspaces of V such that*

$$V = U_1 + \cdots + U_m.$$

Then the following are equivalent:

(i) $V = U_1 \oplus \cdots \oplus U_m$;

(ii) *if* $u_1 \in U_1, \dots, u_m \in U_m$ *and*

$$u_1 + \cdots + u_m = 0,$$

then

$$u_1 = \cdots = u_m = 0.$$

Professional Standard Proof.
TODO: professional standard proof for condition-for-direct-sum. □

Detailed Learning Proof.
TODO: detailed learning proof for condition-for-direct-sum. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Direct Sum of Two Subspaces)**.** *Let U and W be subspaces of a vector space V. Then*

$$U + W = U \oplus W$$

if and only if

$$U \cap W = \{0\}.$$

Professional Standard Proof.
TODO: professional standard proof for direct-sum-of-two-subspaces. □

Detailed Learning Proof.
TODO: detailed learning proof for direct-sum-of-two-subspaces. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Capstone

Chapter 5

Lattice Order

SERIES	VOLUME VI	BOOK III	CHAPTER 5
From Cantor to Ito	Algebra	Lattice and Order Theory	Lattice Order

5.1 Breadcrumb

Why Lattice Theory Gets Its Own Home

Remark (Why lattice theory gets its own home). Lattice theory begins with a simple observation: in many ordered structures, two elements have a natural least upper bound and a natural greatest lower bound. Once those operations exist everywhere, order starts behaving like algebra.

Remark (Where the first pass already lives). The first introduction to lattices is already in the lattice section of the order notes. That is the right place for the first encounter, because it grows directly out of partial orders, suprema, and infima.

Remark (Why return later). Once lattices stop being examples and start becoming the main character, the story gets richer. A fuller lattice-theory chapter will naturally gather topics such as:

- distributive, modular, and Boolean lattices;
- complete lattices and closure operators;
- sublattices, ideals, and filters;
- Galois connections and adjoint situations;
- fixed-point theorems such as Knaster–Tarski;
- the lattice viewpoint behind topologies, sigma-algebras, and domain theory.

Remark (Why this matters for the rest of the curriculum)*.* This chapter is a breadcrumb because lattice order quietly reappears all over the place. Power sets form Boolean lattices, sigma-algebras behave like closure-stable lattice structures, open sets form a join-rich order, and fixed-point methods show up in logic, measure, computation, and analysis. Giving lattice order its own future home makes those later echoes easier to recognize.

5.2 Ordered Sets

5.2.1 Ordered Sets

Remark (Toolkit: Ordered Sets)*.* Order is a relation, not a property. A set carries no intrinsic ordering; an order is an additional structure imposed on the set by specifying which pairs of elements stand in the ordering relation. The same underlying set can support many distinct orders simultaneously. The atomic notions developed here are:

- the **partial order** — the minimal framework in which order makes sense;
- the **strict partial order** — the asymmetric companion to the partial order, and the native language of open-ball conditions and limits;
- the **total order** — the specialisation in which every pair of elements is comparable.

Definition (Partial Order)

Definition 5.2.1 (Partial Order)**.** A relation $\le$ on a set S is a *partial order* if it satisfies, for all $a, b, c \in S$:

1. **Reflexivity:** $a \le a$.
2. **Anti-symmetry:** $a \le b \land b \le a \implies a = b$.
3. **Transitivity:** $a \le b \land b \le c \implies a \le c$.

The pair $(S, \le)$ is called a *partially ordered set*, or *poset*.

Remark (Standard quantified statement)*.*

$$\mathrm{PartialOrder}(S, \le) \;:=\; \big(\forall a \in S,\ a \le a\big) \land \big(\forall a, b \in S,\ a \le b \land b \le a \Rightarrow a = b\big) \land \big(\forall a, b, c \in S,\ a \le b \land b \le c$$

Remark (Negated quantified statement)*.*

$$\neg\,\mathrm{PartialOrder}(S, \le) \;:=\; \big(\exists a \in S,\ a \not\le a\big) \lor \big(\exists a, b \in S,\ a \le b \land b \le a \land a \ne b\big) \lor \big(\exists a, b, c \in S,\ a \le b \land b \le c \land$$

Remark (Interpretation)*.* Order is not an intrinsic quality of a set's elements — it is an external binary relation. The three axioms together enforce that the relation be a consistent, cycle-free, directed comparison. Reflexivity says every element is at least as large as itself. Anti-symmetry is the formal engine behind squeeze arguments: if $a \le b$ and $b \le a$ then a and b cannot be distinct. Transitivity ensures that the ordering chains: comparisons can be composed without breaking consistency. The same set of objects can carry many distinct partial orders simultaneously — standard, reverse, or lexicographic — each defining a different relational geometry.

Remark (Dependencies).

- Partial Order

Definition

Definition 5.2.2 (Strict Partial Order)**.** A relation $<$ on a set S is a *strict partial order* if it satisfies, for all $a, b, c \in S$:

1. **Irreflexivity:** $\neg(a < a)$.
2. **Asymmetry:** $a < b \implies \neg(b < a)$.
3. **Transitivity:** $a < b \wedge b < c \implies a < c$.

Remark (Standard quantified statement).

$$\mathrm{StrictPartialOrder}(S, <) \;\coloneqq\; \big(\forall a \in S,\ \neg(a < a)\big) \wedge \big(\forall a, b \in S,\ a < b \Rightarrow \neg(b < a)\big) \wedge \big(\forall a, b, c \in S,\ a < b \wedge b$$

Remark (Interpretation). The strict order is the companion of the non-strict order, eliminating reflexivity and replacing anti-symmetry with the stronger asymmetry. Every strict partial order determines a partial order by $a \leq b \iff (a < b \vee a = b)$, and every partial order determines a strict partial order by $a < b \iff (a \leq b \wedge a \neq b)$. In metric spaces and analysis, strict orders are the natural language of open conditions: open balls, open intervals, and limit definitions all use $<$ rather than $\leq$ to exclude boundary contact.

Remark (Dependencies).

- Partial Order

Definition (Total Order)

Definition 5.2.3 (Total Order)**.** A partial order $\leq$ on a set S is a *total order* (or *linear order*) if it additionally satisfies the *law of trichotomy*: for all $a, b \in S$, exactly one of

$$a < b, \qquad a = b, \qquad b < a$$

holds. The pair $(S, \leq)$ is called a *totally ordered set* or a *chain*.

Remark (Standard quantified statement).

$$\mathrm{TotalOrder}(S, \leq) \;\coloneqq\; \mathrm{PartialOrder}(S, \leq) \;\wedge\; \forall a, b \in S,\ (a < b) \oplus (a = b) \oplus (b < a),$$

where $\oplus$ denotes exclusive or. Equivalently, comparability suffices: $\forall a, b \in S,\ a \leq b \vee b \leq a$.

Remark (Interpretation). A partial order permits elements to be incomparable: it is possible that neither $a \leq b$ nor $b \leq a$. A total order removes this possibility entirely — every pair of elements is forced to be comparable. The rational numbers $\mathbb{Q}$ and real numbers $\mathbb{R}$ under their standard orderings are totally ordered fields. By contrast, the power set $\mathcal{P}(X)$ of any set X with more than one element, ordered by inclusion, is typically not a total order: the singletons $\{a\}$ and $\{b\}$ with $a \neq b$ are incomparable.

Remark (Dependencies).

Partial order, strict partial order.

5.2.2 Chains and Antichains

Remark (Toolkit: Chains and Antichains). A poset can contain subsets whose elements are either completely aligned or entirely incomparable. These two extremes are formalised as chains and antichains. They measure the "linearity" and "width" of the order structure respectively.

Definition (Chain)

Definition 5.2.4 (Chain)**.** Let $(S, \leq)$ be a partially ordered set. A subset $C \subseteq S$ is a *chain* if every pair of elements in C is comparable:

$$\forall x, y \in C, \quad x \leq y \ \vee \ y \leq x.$$

Equivalently, the restriction of $\leq$ to C is a total order on C.

Remark (Standard quantified statement).

$$\mathrm{Chain}(C, S, \leq) \ \coloneqq \ C \subseteq S \ \wedge \ \forall x, y \in C, \ x \leq y \vee y \leq x.$$

Remark (Negated quantified statement).

$$\neg\,\mathrm{Chain}(C, S, \leq) \ \coloneqq \ \exists x, y \in C, \ x \not\leq y \wedge y \not\leq x.$$

Remark (Interpretation). A chain is a subset in which the partial order behaves as a total order: no two elements are left incomparable. The entire real line $\mathbb{R}$ under its standard ordering is itself a chain. Any finite totally ordered set — a ranking, a sequence of nested intervals, a linearly ordered list of stages — is a chain. Chains model the "ladder" structure of order: there is a clear direction from smaller to larger, and the path from any element to any other is unambiguous.

Remark (Dependencies).

Partial order, total order.

Definition (Antichain)

Definition 5.2.5 (Antichain)**.** Let $(S, \leq)$ be a partially ordered set. A subset $A \subseteq S$ is an *antichain* if no two distinct elements of A are comparable:

$$\forall x, y \in A, \quad x \neq y \Longrightarrow \neg(x \leq y) \wedge \neg(y \leq x).$$

Remark (Standard quantified statement).

$$\mathrm{Antichain}(A, S, \leq) \;:=\; A \subseteq S \;\wedge\; \forall x, y \in A,\; x \neq y \Rightarrow x \not\leq y \wedge y \not\leq x.$$

Remark (Negated quantified statement).

$$\neg\, \mathrm{Antichain}(A, S, \leq) \;:=\; \exists x, y \in A,\; x \neq y \wedge (x \leq y \vee y \leq x).$$

Remark (Interpretation). An antichain is a subset in which the partial order makes no comparisons at all: every pair of distinct elements is incomparable. In the power set $\mathcal{P}(X)$ ordered by inclusion, the collection of all subsets of a fixed cardinality k forms an antichain — no subset of size k contains another. Antichains measure the "width" of a poset: Dilworth's theorem states that the minimum number of chains needed to cover a finite poset equals the maximum size of an antichain. In higher-dimensional structures such as product orders, large antichains arise naturally and are the primary tool for understanding the lateral spread of the order.

Remark (Dependencies).

Partial order.

5.2.3 Partial and Total Maps

Remark (Toolkit: Partial and Total Maps). Before studying maps that preserve order structure, the foundational function types are recorded: a partial map need not be defined everywhere, while a total map is defined on every input. These are the base layer upon which the order-theoretic notions rest.

Definition (Partial Map)

Definition 5.2.6 (Partial Map)**.** Let A and B be sets. A *partial map* $f : A \rightharpoonup B$ is a relation $f \subseteq A \times B$ such that every element of A has at most one image:

$$\forall x \in A,\; \forall y_1, y_2 \in B, \quad (x, y_1) \in f \;\wedge\; (x, y_2) \in f \implies y_1 = y_2.$$

Remark (Standard quantified statement).

$$\mathrm{PartialMap}(f, A, B) \;:=\; f \subseteq A \times B \wedge \forall x \in A,\; \forall y_1, y_2 \in B,\; (x, y_1) \in f \wedge (x, y_2) \in f \Rightarrow y_1 = y_2.$$

Remark (Interpretation). A partial map assigns to each element of A at most one element of B. Its domain of definition $\mathrm{dom}(f) = \{x \in A : \exists y \in B,\, (x, y) \in f\}$ may be a proper subset of A. Partial maps arise in order theory when a function is well-defined only on elements satisfying some order-theoretic condition — for example, on elements that admit an upper bound.

Definition

Definition 5.2.7 (Total Map)**.** A partial map $f : A \rightharpoonup B$ is a *total map* if it is defined on every element of A:

$$\forall x \in A,\; \exists y \in B, \quad (x, y) \in f.$$

A total map is written $f : A \to B$ and is precisely a function in the usual sense.

Remark (Standard quantified statement).

$$\mathrm{TotalMap}(f, A, B) \;\coloneqq\; \mathrm{PartialMap}(f, A, B) \;\wedge\; \forall x \in A,\; \exists y \in B,\; (x, y) \in f.$$

Remark (Dependencies).

Partial map, Function.

5.2.4 Maps Between Ordered Sets

Remark (Toolkit: Maps Between Ordered Sets). The structure-preserving maps between posets form a hierarchy of increasing strength. An order-preserving map carries the direction of the order forward. An order-embedding carries it in both directions. An order-isomorphism is a surjective order-embedding — a perfect structural identification between two posets.

Definition (Order-Preserving Map)

Definition 5.2.8 (Order-Preserving Map)**.** Let $(P, \leq_P)$ and $(Q, \leq_Q)$ be partially ordered sets. A map $\varphi : P \to Q$ is *order-preserving* (or *monotone*) if

$$\forall x, y \in P, \quad x \leq_P y \implies \varphi(x) \leq_Q \varphi(y).$$

Remark (Standard quantified statement).

$$\mathrm{OrderPreserving}(\varphi, P, Q) \;\coloneqq\; \forall x, y \in P,\; x \leq_P y \Rightarrow \varphi(x) \leq_Q \varphi(y).$$

Remark (Negated quantified statement).

$$\neg\, \mathrm{OrderPreserving}(\varphi, P, Q) \;\coloneqq\; \exists x, y \in P,\; x \leq_P y \wedge \varphi(x) \not\leq_Q \varphi(y).$$

Remark (Interpretation). An order-preserving map carries the direction of the order from P into Q: whenever x is at or below y in P, $\varphi(x)$ is at or below $\varphi(y)$ in Q. The implication is one-directional — order-preserving maps need not reflect the order back. Two order-preserving maps compose to an order-preserving map, making order-preserving maps the morphisms of the category of posets.

Remark (Dependencies).

Partial order.

Definition (Order-Embedding)

Definition 5.2.9 (Order-Embedding)**.** Let $(P, \leq_P)$ and $(Q, \leq_Q)$ be partially ordered sets. A map $\varphi : P \to Q$ is an *order-embedding* if

$$\forall x, y \in P, \quad x \leq_P y \iff \varphi(x) \leq_Q \varphi(y).$$

When φ is an order-embedding, one writes $\varphi : P \hookrightarrow Q$.

Remark (Standard quantified statement).

$$\mathrm{OrderEmbedding}(\varphi, P, Q) \;\coloneqq\; \forall x, y \in P,\; x \leq_P y \iff \varphi(x) \leq_Q \varphi(y).$$

Remark (Interpretation). An order-embedding both preserves and reflects the order: the order structure of P is faithfully copied into Q without distortion. Unlike mere preservation, reflection means that order-relations between images in Q genuinely originate from order-relations in P. Every order-embedding is injective (Lemma 1.2.11) and identifies P order-isomorphically with its image in Q.

Remark (Dependencies).

Order-preserving map.

Definition (Order-Isomorphism)

Definition 5.2.10 (Order-Isomorphism)**.** An order-embedding $\varphi : P \to Q$ that is surjective is an *order-isomorphism*. When an order-isomorphism exists, P and Q are said to be *order-isomorphic*, written $P \cong Q$.

Remark (Standard quantified statement).

$$\mathrm{OrderIso}(\varphi, P, Q) \;\coloneqq\; \mathrm{OrderEmbedding}(\varphi, P, Q) \;\wedge\; \forall q \in Q,\; \exists p \in P,\; \varphi(p) = q.$$

Remark (Interpretation). An order-isomorphism is a bijection that perfectly translates the order of P into Q and back. Two order-isomorphic posets are order-theoretically identical: any statement about P expressed purely in terms of $\leq_P$ translates to an equally valid statement about Q under φ. Order-isomorphism is the correct notion of equality for posets.

Remark (Dependencies).

Order-embedding.

Lemma

Lemma 5.2.11 (Every Order-Embedding Is Injective)**.** *Let $(P, \leq_P)$ and $(Q, \leq_Q)$ be partially ordered sets, and let $\varphi : P \to Q$ be an order-embedding. Then φ is injective.* *Go to proof.*

Remark (Standard quantified statement).

$$\mathrm{OrderEmbedding}(\varphi, P, Q) \implies \forall x, y \in P,\; \varphi(x) = \varphi(y) \Rightarrow x = y.$$

Remark (Interpretation). Injectivity follows automatically from the biconditional in the embedding definition. If $\varphi(x) = \varphi(y)$, the image is simultaneously $\leq$ and $\geq$ itself in Q; order-reflection pulls this back to P, and anti-symmetry then forces $x = y$.

Remark (Dependencies).

Order-embedding, anti-symmetry of partial orders.

5.2.5 The Covering Relation

Remark (Toolkit: The Covering Relation). The covering relation identifies the irreducible steps of a partial order: y covers x when $x < y$ and there is nothing strictly between them. In a finite poset the entire order is determined by its covering relation, which is encoded exactly by the Hasse diagram.

Definition (Covering Relation)

Definition 5.2.12 (Covering Relation)**.** Let $(P, \leq)$ be a partially ordered set and let $x, y \in P$ with $x < y$. The element y *covers* x, written $x \prec y$, if there is no element of P strictly between them:

$$x \prec y \;\coloneqq\; x < y \;\wedge\; \neg\big(\exists z \in P,\; x < z < y\big).$$

Remark (Standard quantified statement).

$$x \prec y \;\coloneqq\; x < y \;\wedge\; \forall z \in P,\; x < z \leq y \Longrightarrow z = y.$$

Remark (Negated quantified statement).

$$\neg(x \prec y) \;\coloneqq\; x \not< y \;\vee\; \exists z \in P,\; x < z < y.$$

Remark (Interpretation). The covering relation captures the direct steps of the order — the comparisons that cannot be decomposed further. In a finite poset every strict comparison $x < y$ can be resolved into a finite chain of covers $x = x_0 \prec x_1 \prec \cdots \prec x_n = y$, so the covering relation alone determines the full order. The Hasse diagram of a finite poset is exactly the directed graph of the covering relation. In a dense order such as $\mathbb{Q}$ or $\mathbb{R}$, no element covers any other: between any two elements there is always a third.

Remark (Dependencies).

Partial order, strict partial order.

Lemma

Lemma 5.2.13 (Characterizations of Order-Isomorphisms for Finite Posets)**.** *Let $(P, \leq_P)$ and $(Q, \leq_Q)$ be finite partially ordered sets and let $\varphi : P \to Q$ be a bijection. The following are equivalent:*

1. *φ is an order-isomorphism.*
2. *For all $x, y \in P$: $x <_P y \iff \varphi(x) <_Q \varphi(y)$.*
3. *For all $x, y \in P$: $x \prec_P y \iff \varphi(x) \prec_Q \varphi(y)$.*

Go to proof.

Remark (Standard quantified statement).

$$\mathrm{OrderIso}(\varphi, P, Q) \iff \big(\forall x, y \in P,\; x <_P y \Leftrightarrow \varphi(x) <_Q \varphi(y)\big) \iff \big(\forall x, y \in P,\; x \prec_P y \Leftrightarrow \varphi(x) \prec_Q \varphi(y)\big)$$

Remark (Interpretation). In a finite poset the order relation, the strict order, and the covering relation each determine the others completely. This lemma shows that an order-isomorphism can be detected at any one of these three levels. The covering characterisation is the most useful in practice: since Hasse diagrams encode exactly the covering relation, two finite posets are order-isomorphic precisely when their diagrams are isomorphic as directed graphs.

Remark (Dependencies).

Order-isomorphism, covering relation.

Proposition

Proposition 5.2.14 (Hasse Diagram Characterization for Finite Posets)**.** *Let* $(P, \leq_P)$ *and* $(Q, \leq_Q)$ *be finite partially ordered sets. Then* P *and* Q *are order-isomorphic if and only if their Hasse diagrams are isomorphic as directed graphs — that is, identical up to relabeling of elements.* *Go to proof.*

Remark (Standard quantified statement).

$$P \cong Q \iff \exists \varphi : P \xrightarrow{\sim} Q,\ \forall x, y \in P,\ x \prec_P y \Leftrightarrow \varphi(x) \prec_Q \varphi(y).$$

Remark (Interpretation). The Hasse diagram is a complete invariant of the order structure of a finite poset. Checking order-isomorphism between finite posets reduces to a graph isomorphism problem on the covering graphs — two finite posets with non-isomorphic Hasse diagrams cannot be order-isomorphic.

Remark (Dependencies).

Characterizations of order-isomorphisms for finite posets, covering relation.

5.2.6 Duality

Remark (Toolkit: Duality). Every partially ordered set has a dual obtained by reversing all order relations. Any statement that holds in every poset therefore holds in every dual poset, which means its order-reversed version also holds universally. This is the Duality Principle — the formal engine behind phrases like "by duality, the corresponding result for infima follows."

Definition (Dual Ordered Set)

Definition 5.2.15 (Dual Ordered Set)**.** Let $(P, \leq)$ be a partially ordered set. The *dual* (or *opposite*) of P, denoted P^{op}, is the set P equipped with the reversed order:

$$x \leq_{P^{\mathrm{op}}} y \iff y \leq_P x.$$

Remark (Standard quantified statement).

$$x \leq_{P^{\mathrm{op}}} y \coloneqq y \leq_P x.$$

Remark (Interpretation). Dualising a poset exchanges the roles of upper and lower: every upper bound becomes a lower bound, every supremum becomes an infimum, and every maximal element becomes a minimal element. In a finite poset the Hasse diagram of P^{op} is obtained from that of P by turning it upside down. The dual of a total order is again a total order.

Remark (Dependencies).

Partial order.

Proposition

Proposition 5.2.16 (Duality Principle)**.** *Let Φ be a statement about partially ordered sets, expressed purely in terms of $\leq$, that holds in every partially ordered set. Then the dual statement Φ^{op}, obtained by replacing every occurrence of $\leq$ with $\geq$, also holds in every partially ordered set.* *Go to proof.*

Remark (Standard quantified statement). Every universally valid statement about partially ordered sets has a universally valid order-dual statement.

Remark (Interpretation). The Duality Principle eliminates the need to re-prove results whose dual is already established. Any theorem about upper bounds, suprema, maximal elements, or top elements automatically yields a theorem about lower bounds, infima, minimal elements, or bottom elements by applying the principle to P^{op}.

Remark (Dependencies).

Dual ordered set.

5.2.7 Extremal Elements

Remark (Toolkit: Extremal Elements). A poset may contain elements that sit at the boundary of the order. There are two distinct senses of “extreme”: an element can be extreme *locally* (nothing in the poset exceeds it, though it may be incomparable with some elements) or extreme *globally* (it dominates or is dominated by every element). The definitions below make these distinctions precise, proceeding from global extremes of the whole poset, through local extremes of a subset, to the proposition that finite posets always contain the local kind.

Definition (Bottom Element)

Definition 5.2.17 (Bottom Element)**.** Let $(P, \leq)$ be a partially ordered set. An element $\bot \in P$ is a *bottom element* (or *least element*) of P if

$$\forall x \in P, \quad \bot \leq x.$$

If a bottom element exists it is unique; it is denoted $\bot_P$ or simply $\bot$.

Remark (Standard quantified statement).

$$\text{Bottom}(\bot, P, \leq) \;\coloneqq\; \bot \in P \;\wedge\; \forall x \in P,\; \bot \leq x.$$

Remark (Interpretation). A bottom element is below every element of the poset without exception. In $(\mathcal{P}(X), \subseteq)$ the bottom element is $\varnothing$. The natural numbers have a bottom element; the integers do not. An antichain with more than one element has no bottom element, since no member is below any other.

Remark (Dependencies).

Partial order.

Definition (Top Element)

Definition 5.2.18 (Top Element)**.** Let $(P, \leq)$ be a partially ordered set. An element $\top \in P$ is a *top element* (or *greatest element*) of P if

$$\forall x \in P, \quad x \leq \top.$$

If a top element exists it is unique; it is denoted $\top_P$ or simply $\top$.

Remark (Standard quantified statement).

$$\mathrm{Top}(\top, P, \leq) \;\coloneqq\; \top \in P \;\wedge\; \forall x \in P,\; x \leq \top.$$

Remark (Interpretation). Dual to the bottom element via the Duality Principle. In $(\mathcal{P}(X), \subseteq)$ the top element is X itself.

Remark (Dependencies).

Bottom element, duality principle.

Definition

Definition 5.2.19 (Maximal Element)**.** Let $(P, \leq)$ be a partially ordered set and let $Q \subseteq P$. An element $a \in Q$ is *maximal in* Q if no element of Q strictly exceeds it:

$$\forall x \in Q, \quad a \leq x \implies a = x.$$

The set of maximal elements of Q is denoted $\mathrm{Max}(Q)$.

Remark (Standard quantified statement).

$$\mathrm{Maximal}(a, Q, \leq) \;\coloneqq\; a \in Q \;\wedge\; \forall x \in Q,\; a \leq x \Rightarrow a = x.$$

Remark (Negated quantified statement).

$$\neg\,\mathrm{Maximal}(a, Q, \leq) \;\coloneqq\; \exists x \in Q,\; a < x,$$

that is, some element of Q strictly exceeds a.

Remark (Interpretation). A maximal element asserts only that nothing in Q strictly dominates it. It need not be comparable with every element of Q, so a poset can have multiple maximal elements simultaneously. In domain theory, maximal elements correspond to fully-specified or total objects.

Remark (Dependencies).

Partial order.

Definition

Definition 5.2.20 (Minimal Element)**.** Let $(P, \leq)$ be a partially ordered set and let $Q \subseteq P$. An element $a \in Q$ is *minimal in* Q if no element of Q is strictly below it:

$$\forall x \in Q, \quad x \leq a \implies x = a.$$

Remark (Standard quantified statement)*.*

$$\text{Minimal}(a, Q, \leq) \;\coloneqq\; a \in Q \;\wedge\; \forall x \in Q,\; x \leq a \Rightarrow x = a.$$

Remark (Interpretation)*.* Dual to maximal element via the Duality Principle.

Remark (Dependencies)*.*

Maximal element, duality principle.

Definition

Definition 5.2.21 (Greatest Element)**.** Let $(P, \leq)$ be a partially ordered set and let $Q \subseteq P$. An element $m \in Q$ is the *greatest element* (or *maximum*) of Q if it dominates every element of Q:

$$\forall x \in Q, \quad x \leq m.$$

If it exists it is unique; it is denoted $\max Q$.

Remark (Standard quantified statement)*.*

$$\text{Greatest}(m, Q, \leq) \;\coloneqq\; m \in Q \;\wedge\; \forall x \in Q,\; x \leq m.$$

Remark (Interpretation)*.* A greatest element is a maximal element that is additionally comparable with every element of Q. When $\max Q$ exists, $\text{Max}(Q) = \{\max Q\}$, but the converse fails: a unique maximal element need not dominate every element.

Remark (Dependencies)*.*

Maximal element.

Definition

Definition 5.2.22 (Least Element)**.** Let $(P, \leq)$ be a partially ordered set and let $Q \subseteq P$. An element $m \in Q$ is the *least element* (or *minimum*) of Q if it lies below every element of Q:

$$\forall x \in Q, \quad m \leq x.$$

If it exists it is unique; it is denoted $\min Q$.

Remark (Standard quantified statement)*.*

$$\text{Least}(m, Q, \leq) \;\coloneqq\; m \in Q \;\wedge\; \forall x \in Q,\; m \leq x.$$

Remark (Interpretation)*.* Dual to greatest element.

Remark (Dependencies).

Greatest element, duality principle.

Proposition

Proposition 5.2.23 (Every Nonempty Finite Poset Has a Maximal Element)**.** *Let* $(P, \leq)$ *be a finite partially ordered set. Then every nonempty subset* $Q \subseteq P$ *has at least one maximal element. Furthermore, for every* $x \in P$ *there exists* $y \in \mathrm{Max}(P)$ *with* $x \leq y$*.* *Go to proof.*

Remark (Standard quantified statement).

$$P \text{ finite},\ Q \subseteq P,\ Q \neq \varnothing \implies \mathrm{Max}(Q) \neq \varnothing.$$

$$\forall x \in P,\ \exists y \in \mathrm{Max}(P),\ x \leq y.$$

Remark (Interpretation). Finiteness is essential. The collection of all finite subsets of $\mathbb{N}$, ordered by inclusion, is infinite and has no maximal element. In a finite poset one can always ascend from any element to a ceiling. This property is the finite shadow of Zorn's Lemma and underlies ascending chain arguments throughout algebra and order theory.

Remark (Dependencies).

Maximal element, partial order.

5.2.8 Lifting and Co-Lifting

Remark (Toolkit: Lifting and Co-Lifting). A poset that lacks a bottom or top element can be given one by adjoining a new universal bound. Lifting adjoins a bottom; co-lifting adjoins a top. These constructions are dual to each other and are used throughout domain theory, where a bottom element represents an undefined state.

Definition (Lifting)

Definition 5.2.24 (Lifting)**.** Let $(P, \leq)$ be a partially ordered set. The *lifting* of P is the poset $P_\perp := P \cup \{\perp\}$, where $\perp \notin P$, with the order:

$$\perp \leq x \quad \text{for all } x \in P_\perp, \qquad x \leq_{P_\perp} y \iff x \leq_P y \quad \text{for all } x, y \in P.$$

Thus $P_\perp$ is P with a new bottom element $\perp$ adjoined.

Remark (Standard quantified statement).

$$P_\perp = P \cup \{\perp\}, \quad \perp \notin P, \quad \forall x \in P_\perp,\ \perp \leq x, \quad \forall x, y \in P,\ x \leq_{P_\perp} y \Leftrightarrow x \leq_P y.$$

Remark (Interpretation). Lifting adjoins a universal lower bound to a poset that may not have one. In domain theory $\perp$ represents an undefined, divergent, or unspecified state. Any set S viewed as an antichain can be lifted to the *flat poset* $S_\perp$, in which all elements of S are incomparable above $\perp$.

Remark (Dependencies).

Partial order, Union, bottom element.

Definition

Definition 5.2.25 (Co-Lifting)**.** Let $(P, \leq)$ be a partially ordered set. The *co-lifting* of P is the poset $P^{\top} := P \cup \{\top\}$, where $\top \notin P$, with the order:

$$x \leq \top \quad \text{for all } x \in P^{\top}, \qquad x \leq_{P^{\top}} y \iff x \leq_P y \quad \text{for all } x, y \in P.$$

Thus $P^{\top}$ is P with a new top element $\top$ adjoined.

Remark (Standard quantified statement).

$$P^{\top} = P \cup \{\top\}, \quad \top \notin P, \quad \forall x \in P^{\top},\ x \leq \top, \quad \forall x, y \in P,\ x \leq_{P^{\top}} y \Leftrightarrow x \leq_P y.$$

Remark (Interpretation). Dual to lifting via the Duality Principle: co-lifting adjoins a universal upper bound. The co-lifting of an antichain produces a flat poset in which all elements are incomparable below $\top$.

Remark (Dependencies).

Lifting, top element, duality principle.

5.2.9 Consistency

Remark (Toolkit: Consistency). Two elements of a poset are consistent when they share a common upper bound — when they can be jointly extended to some element above both of them. In lattice-theoretic settings this is automatic; in general posets it is a genuine condition.

Definition

Definition 5.2.26 (Consistency)**.** Let $(S, \leq)$ be a partially ordered set. Elements $x, y \in S$ are *consistent* (or *compatible*) if they admit a common upper bound:

$$\exists z \in S, \quad x \leq z \ \wedge \ y \leq z.$$

Remark (Standard quantified statement).

$$\mathrm{Consistent}(x, y, S, \leq) \ := \ \exists z \in S,\ x \leq z \wedge y \leq z.$$

Remark (Negated quantified statement).

$$\neg\,\mathrm{Consistent}(x, y, S, \leq) \ := \ \forall z \in S,\ x \not\leq z \vee y \not\leq z.$$

Remark (Interpretation). Consistency asserts that x and y are compatible — there is some element of S above both. In a lattice where binary joins always exist, consistency is automatic: take $z = x \vee y$. In a general poset the condition is nontrivial and may fail. In domain theory, consistency captures the idea that two partial computations describe information that can be combined without contradiction.

Remark (Dependencies).
Partial order.

5.2.10 Down-Sets and Up-Sets

Remark (Toolkit: Down-Sets and Up-Sets). A down-set is a subset of a poset that is closed downward: if an element belongs to it, everything below that element belongs to it too. An up-set is the dual notion — closed upward. Down-sets are the order ideals of lattice theory; up-sets are the order filters. The collection of all down-sets of a poset, ordered by inclusion, is itself an important poset.

Definition (Down-Set)

Definition 5.2.27 (Down-Set)**.** Let $(P, \leq)$ be a partially ordered set. A subset $Q \subseteq P$ is a *down-set* (or *decreasing set*, or *order ideal*) if it is closed downward:

$$\forall x \in Q,\ \forall y \in P, \quad y \leq x \implies y \in Q.$$

Remark (Standard quantified statement).

$$\mathrm{DownSet}(Q, P, \leq) \ \coloneqq\ Q \subseteq P \ \wedge\ \forall x \in Q,\ \forall y \in P,\ y \leq x \Rightarrow y \in Q.$$

Remark (Negated quantified statement).

$$\neg\,\mathrm{DownSet}(Q, P, \leq) \ \coloneqq\ \exists x \in Q,\ \exists y \in P,\ y \leq x \wedge y \notin Q.$$

Remark (Interpretation). A down-set is a downward-closed region of the poset: once an element is included, the entire lower section below it is forced in. Down-sets generalise downward-closed intervals. The collection of all down-sets of P, denoted $\mathcal{O}(P)$ and ordered by inclusion, is itself a poset and in the finite case a distributive lattice. Birkhoff's representation theorem states that every finite distributive lattice arises this way.

Remark (Dependencies).
Partial order.

Definition (Up-Set)

Definition 5.2.28 (Up-Set)**.** Let $(P, \leq)$ be a partially ordered set. A subset $Q \subseteq P$ is an *up-set* (or *increasing set*, or *order filter*) if it is closed upward:

$$\forall x \in Q,\ \forall y \in P, \quad x \leq y \implies y \in Q.$$

Remark (Standard quantified statement).

$$\mathrm{UpSet}(Q, P, \leq) \ \coloneqq\ Q \subseteq P \ \wedge\ \forall x \in Q,\ \forall y \in P,\ x \leq y \Rightarrow y \in Q.$$

Remark (Negated quantified statement).

$$\neg\,\mathrm{UpSet}(Q, P, \leq) \ \coloneqq\ \exists x \in Q,\ \exists y \in P,\ x \leq y \wedge y \notin Q.$$

Remark (Interpretation). Dual to down-set via the Duality Principle: the up-sets of P are exactly the down-sets of P^{op}. In topology, filters are special up-sets closed under finite intersection and satisfying a directedness condition. The complement of a down-set is an up-set, and vice versa.

Remark (Dependencies).

Down-set, duality principle.

5.2.11 Families of Sets

Remark (Toolkit: Families of Sets). The power set of any set is naturally ordered by inclusion. Every collection of subsets — whether arising from algebra, topology, or logic — inherits this order. The key observation is that subsets correspond exactly to predicates via their characteristic functions, and that this correspondence is an order-isomorphism.

Remark (Families as ordered sets). Let X be a set. The *power set* $\mathcal{P}(X)$, consisting of all subsets of X, is ordered by inclusion: for $A, B \in \mathcal{P}(X)$,

$$A \leq B \iff A \subseteq B.$$

Any subcollection of $\mathcal{P}(X)$ inherits this order. Such a collection is called a *family of sets*.

Families of sets arise naturally when X carries additional structure. If X is a group G, the set $\mathrm{Sub}(G)$ of all subgroups and the set $\mathrm{NSub}(G)$ of all normal subgroups are each ordered by inclusion. If X is a vector space V, the set $\mathrm{Sub}(V)$ of all subspaces is ordered by inclusion. If X is a ring R, the sets of all subrings and of all ideals of R are similarly ordered. If $(X, \mathcal{T})$ is a topological space, the collections of open sets, closed sets, and clopen sets are each families of sets ordered by inclusion.

A particularly important example is the family $\mathcal{O}(P)$ of all down-sets of a partially ordered set P, ordered by inclusion.

Remark (Predicate formulation). The ordered set $(\mathcal{P}(X), \subseteq)$ admits an equivalent formulation in terms of predicates. A *predicate* on X is a function

$$p : X \to \{T, F\},$$

where T and F denote truth values. Let $\mathcal{P}^*(X)$ denote the set of all predicates on X, ordered by implication:

$$p \leq q \iff \{x \in X : p(x) = T\} \subseteq \{x \in X : q(x) = T\}.$$

Define the map

$$\Phi : \mathcal{P}^*(X) \to \mathcal{P}(X), \qquad \Phi(p) := \{x \in X : p(x) = T\}.$$

Proposition

Proposition 5.2.29 (Predicates and Power Sets Are Order-Isomorphic). *The map $\Phi : (\mathcal{P}^*(X), \leq) \to (\mathcal{P}(X), \subseteq)$ is an order-isomorphism.* *Go to proof.*

Remark (Standard quantified statement).

$$\mathrm{OrderIso}(\Phi, \mathcal{P}^*(X), \mathcal{P}(X)).$$

Remark (Interpretation). Every subset of X can be represented as the truth set of a predicate, and every predicate determines a subset. The correspondence is bijective and order-preserving in both directions: $p \leq q$ if and only if the truth set of p is contained in the truth set of q, which is exactly the inclusion order on $\mathcal{P}(X)$. This identification is fundamental in logic, where sets and their characteristic predicates are treated as interchangeable, and in computer science, where sets are often implemented via their membership tests.

Remark (Dependencies).

Order-isomorphism, partial order.

Proofs

Remark (Return). Return to Theorem

Lemma (Characterizations of Order-Isomorphisms for Finite Posets)**.** *Let* $(P, \leq_P)$ *and* $(Q, \leq_Q)$ *be finite partially ordered sets and let* $\varphi : P \to Q$ *be a bijection. The following are equivalent:*

1. φ *is an order-isomorphism.*
2. *For all* $x, y \in P$*:* $x <_P y \iff \varphi(x) <_Q \varphi(y)$*.*
3. *For all* $x, y \in P$*:* $x \prec_P y \iff \varphi(x) \prec_Q \varphi(y)$*.*

Professional Standard Proof.
TODO: professional standard proof for order-iso-finite-characterizations. □

Detailed Learning Proof.
TODO: detailed learning proof for order-iso-finite-characterizations. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Hasse Diagram Characterization for Finite Posets)**.** *Let* $(P, \leq_P)$ *and* $(Q, \leq_Q)$ *be finite partially ordered sets. Then* P *and* Q *are order-isomorphic if and only if their Hasse diagrams are isomorphic as directed graphs — that is, identical up to relabeling of elements.*

Professional Standard Proof.
TODO: professional standard proof for hasse-diagram-characterization. ☐

Detailed Learning Proof.
TODO: detailed learning proof for hasse-diagram-characterization. ☐

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Duality Principle)**.** *Let Φ be a statement about partially ordered sets, expressed purely in terms of $\leq$, that holds in every partially ordered set. Then the dual statement Φ^{op}, obtained by replacing every occurrence of $\leq$ with $\geq$, also holds in every partially ordered set.*

Professional Standard Proof.
TODO: professional standard proof for duality-principle. □

Detailed Learning Proof.
TODO: detailed learning proof for duality-principle. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Every Nonempty Finite Poset Has a Maximal Element)**.** *Let* $(P, \leq)$ *be a finite partially ordered set. Then every nonempty subset* $Q \subseteq P$ *has at least one maximal element. Furthermore, for every* $x \in P$ *there exists* $y \in \mathrm{Max}(P)$ *with* $x \leq y$*.*

Professional Standard Proof.
TODO: professional standard proof for finite-poset-maximal. □

Detailed Learning Proof.
TODO: detailed learning proof for finite-poset-maximal. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Predicates and Power Sets Are Order-Isomorphic). *The map* $\Phi : (\mathcal{P}^*(X), \leq) \to (\mathcal{P}(X), \subseteq)$ *is an order-isomorphism.*

Professional Standard Proof.
TODO: professional standard proof for predicates-power-sets-iso. □

Detailed Learning Proof.
TODO: detailed learning proof for predicates-power-sets-iso. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Lemma (Every Order-Embedding Is Injective). *Let $(P, \leq_P)$ and $(Q, \leq_Q)$ be partially ordered sets, and let $\varphi : P \to Q$ be an order-embedding. Then φ is injective.*

Professional Standard Proof.
TODO: professional standard proof for order-embedding-injective. □

Detailed Learning Proof.
TODO: detailed learning proof for order-embedding-injective. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Capstone

Bibliography